The Price of Context: Privacy-Utility Trade-offs in Synthetic Counseling Session Generation

Anonymous ACL submission

Abstract

The rise in mental health disorders has increased interest in LLM-based counseling agents. However, general-purpose LLMs underperform on counseling tasks, motivating the use of specialized models trained on sensitive real-world counseling data, which is often unavailable due to privacy constraints. Synthetic counseling session generation offers a potential solution, but the privacy–utility trade-off involved in different private input choices remains underexplored. In this work, we systematically study this trade-off across three privacy levels: SO (symptoms only), CS (symptoms + situation), and FP (symptoms + situation + demographics). We evaluate utility through psychological quality and diversity, and privacy through attribute extraction, membership inference, and linkage attacks. Results show that incorporating situations improves diversity, but substantially increases privacy risks. We further evaluate defenses, including NER-based redaction, demographic coarsening and situation generalization, and two proposed noise-based methods: demographic perturbation and iterative situation reframing. While simple defenses provide limited protection over FP generations, our noise-based methods reduce linkage and membership inference attack effectiveness by 20% each and attribute extraction leakage by 35%, while preserving diversity[1].

1 Introduction

The growing global burden of mental health disorders[2], coupled with a shortage of trained clinicians (Kazdin, 2021), has created an urgent need for scalable counseling solutions. Recent advances in large language models (LLMs) highlight their potential as conversational agents for mental health support (Raile, 2024; Moell, 2024). However, models trained on general-domain data often underperform on counseling tasks compared to those trained on specialized counseling datasets (Zhang et al., 2024; Qiu et al., 2024; Lee et al., 2024).

However, counseling data is highly sensitive, containing mental health information along with Personally Identifiable Information (PII) and quasi-identifiers (e.g., age, occupation) (Mandal et al., 2025c). As a result, access to such datasets remains limited. Existing approaches such as manual anonymization and automated pseudonymization (Tang et al., 2019; Yue and Zhou, 2020) often lack robustness and scalability. Consequently, synthetic data generation has emerged as a promising alternative, aiming to balance utility with privacy.

To mitigate privacy risks, most synthetic counseling session generation approaches avoid directly using private information from real sessions. Prior work varies in generation strategies: some rely on symptom descriptions (De Duro et al., 2025; Chen et al., 2023a), while others construct synthetic patient profiles (Zhezherau and Yanockin, 2024; BN et al., 2025). Other approaches ground generation in anonymized online posts (Qiu et al., 2024; Chen et al., 2023b), counselor or patient notes (Mousavi et al., 2021; Zhang et al., 2024), or crowdsourced profiles (Yao et al., 2022; Lee et al., 2024). A smaller subset uses real data more directly by extracting patient profiles from real sessions (Yin et al., 2025; Wang et al., 2025) or anonymizing and reusing conversations (Liu et al., 2023a; Xu et al., 2025; Qiu and Lan, 2025).

However, synthetic data does not inherently guarantee privacy and can remain vulnerable to attacks when derived from real data (e.g., patient profiles) (Stadler et al., 2022; Sarmin et al., 2025). Conversely, relying solely on synthetic priors (e.g., synthetic profiles) may limit thematic diversity and generalization to real-world settings (Zhang, 2025). This suggests a fundamental trade-off: incorporating richer private context, such as patient back-

[1] Our code is attached in the Appendix.
[2] https://www.who.int/news-room/fact-sheets/detail/mental-disorders

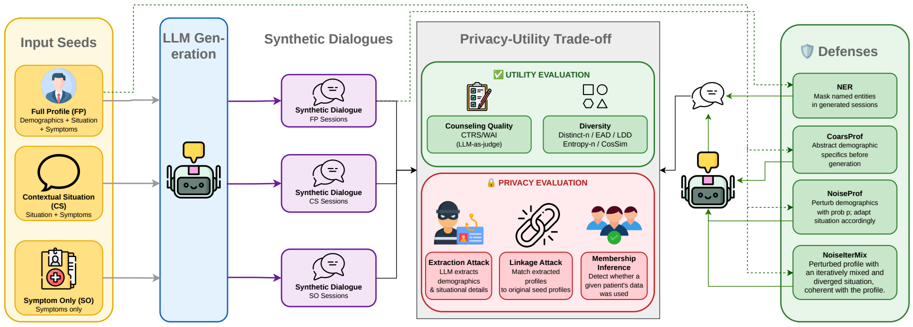

Figure 1: **Overview of our framework for evaluating the privacy–utility trade-off in synthetic counseling session generation.** We vary the amount of private information provided for generation, evaluate utility through counseling quality and diversity, measure privacy risks through attribute extraction, linkage, and membership inference attacks, and study defense mechanisms for improving the trade-off.

ground information, can improve realism and diversity but increases privacy risks. Despite its importance, prior work has not systematically studied how progressively introducing more private information affects this utility-privacy trade-off in synthetic session generation, or whether existing defenses can effectively improve this trade-off.

To address this gap, we systematically evaluate the privacy–utility trade-off in synthetic counseling session generation. We measure utility using counseling quality and diversity. Quality is evaluated using an LLM-as-a-judge framework on established psychological scales, including the Cognitive Therapy Rating Scale (CTRS) (Aarons et al., 2012) and Working Alliance Inventory (WAI) (Horvath and Greenberg, 1989). Diversity is assessed using Distinct-n (Qiu et al., 2024), Expected Adjusted Distinct (EAD) (Liu et al., 2022), Entropy-n (Zhang et al., 2018), Lexical Diversity Density (LDD) (Qiu and Lan, 2025), and cosine similarity between the patient situations from the synthetic sessions (Qiu et al., 2024). For privacy, we adopt a data-holder perspective and measure disclosure risk (Zhang et al., 2022) in the generated synthetic sessions through attribute extraction, linkage, and membership inference attacks.

Based on this evaluation framework, we investigate two research questions: (1) **How does the amount of private information used for synthetic counseling session generation affect the privacy–utility trade-off?** We progressively provide the generation pipeline with (i) symptoms, (ii) symptoms + situation, and (iii) symptoms + situation + demographics. We find that counseling quality remains largely unchanged as private input increases. However, background situations improve diversity while also increasing privacy leakage. (2) **How effective are defenses in improving the trade-off?** We evaluate NER-based redaction, demographic coarsening with situation generalization, and propose two noise-based defenses: noisy demographic perturbation and iterative background situation divergence. While existing defenses provide limited protection, our proposed noise-based defenses reduce membership inference and linkage attack performance by 20% each and attribute leakage by 35% when using full private patient information, while preserving diversity. An overview of the workflow is presented in Figure 1.

2 Related Work

Synthetic Counseling Session Generation. Synthetic counseling sessions have been widely explored for training mental health LLMs (Liu et al., 2023a; Zhang et al., 2024; Qiu et al., 2024; Xu et al., 2025; Lee et al., 2024), with existing approaches differing primarily in the type and amount of private information used for generation. Some methods directly leverage real counseling sessions (Wu et al., 2023; Zhang et al., 2026), while others transform them into QA pairs (Liu et al., 2023a; Xu et al., 2025), use them for masked dialogue reconstruction tasks (Qiu and Lan, 2025), or derive intermediate representations such as patient profiles, counseling goals, symptoms, questionnaires, and psychological graphs (Wang et al., 2025; Yin et al., 2025; De Duro et al., 2025; Chen et al., 2023a; Vu et al., 2025; Mandal et al., 2026). Other approaches incorporate structured clinical information, including counseling notes and structured cases (Mousavi et al., 2021; Zhang et al., 2024; Hu et al., 2026; Xie et al., 2025; Li et al., 2026), or rely on publicly available, crowdsourced, and literature-based resources (Qiu et al., 2024; Chen et al., 2023b; Na, 2024; Chen and Liu, 2025; Su, 2025; Wang et al., 2024; Chen et al., 2025; Zhang, 2025; Nguyen et al., 2026; Mishra et al., 2023; Yao et al., 2022; Maddela et al., 2023; Pan et al., 2026; Zhou et al., 2025). These resources are used either directly for dialogue generation (Xiao et al., 2024) or through derived representations such as profiles (Lee et al., 2024; Mandal et al., 2025b; Lee et al., 2026). Finally, some approaches avoid real data entirely through direct prompting (Cabrera Lozoya et al., 2025), synthetic profiles or case reports (Zhezherau and Yanockin, 2024; Lu et al., 2025), attribute sampling (Ozgun et al., 2025; BN et al., 2025), or structured generation of profiles and stress trajectories (Gou and Liang, 2026). Despite this diversity in using private data, prior work has not systematically quantified how the choice of generation information affects the privacy and utility of synthetic sessions. We address this gap by studying the privacy–utility trade-off across varying levels of information disclosure, including demographic, situational, and symptom information.

Privacy Attacks on Synthetic Data. Synthetic data does not inherently guarantee privacy, as generative models remain vulnerable to attacks (Stadler et al., 2022; Sarmin et al., 2025). Prior work has studied these risks through attacks like reconstruction and membership inference attacks (Shokri et al., 2017; Carlini et al., 2021, 2022). Existing attacks primarily consider white-box access, where the attacker can access to model internals, or black-box access, where only model outputs are accessible (Hayes et al., 2019; Chen et al., 2020; Yang et al., 2023). Some black-box attacks additionally assume access to auxiliary data (Golob et al., 2025; Andrey et al., 2025), often using shadow models to approximate the target model behavior (Houssiau et al., 2022; German et al., 2025). Other approaches exploit properties of the generated data distribution to infer membership (van Breugel et al., 2023; Ward et al., 2025; Mustaquim et al., 2025). More restrictive settings consider attackers with access only to the released synthetic data (Guépin et al., 2023).

For text generation, LLMs are increasingly used to create synthetic data (Long et al., 2024), including counseling dialogues (Mandal et al., 2025a), but LLMs can leak information from training data (Meeus et al., 2025), fine-tuning data (Mireshghallah et al., 2022; Zhu et al., 2024), and in-context examples (Duan et al., 2023; Wen et al., 2024; Xia et al., 2026; Choi et al., 2025; Hou et al., 2025). However, these attacks typically assume direct exposure of private data in training or in-context demonstrations. SynBench (Sun et al., 2025) examines a no-training setting using AugPE (Xie et al., 2024), but focuses on homogeneous generation (using short text to generate its synthetic counterpart). In contrast, we generate counseling sessions from structured profiles derived from private conversations, without exposing raw sessions, resulting in a mismatch between inputs and outputs that remains underexplored.

LLM-based attacks. Recent work highlights increasing privacy risks from LLM-based inference and re-identification (Du et al., 2025b). LLMs can infer sensitive attributes from pseudonymized text, including online posts (Staab et al., 2024a; Yukhymenko et al., 2024; Liu et al., 2025b; Yang et al., 2025), with LLM agents further improving such capabilities (Du et al., 2025a). They also enable re-identification (Nyffenegger et al., 2024; Li, 2026), including of redacted clinical records (Morris et al., 2024), with some approaches leveraging auxiliary social media data to link sanitized medical records (Xin et al., 2025). Recent agent-based deanonymization further demonstrates the scalability of these attacks (Lermen et al., 2026). Building on these advances, we study LLM-based privacy attacks on synthetic counseling sessions, where attackers recover patient attributes and situational details for linkage and membership inference.

3 Synthetic Session Generation with Varying Private Information Inputs

To study the privacy–utility trade-off in synthetic counseling session generation, we define three privacy levels corresponding to progressively increased access to private information. In all settings, we prompt the model to generate multi-turn counseling dialogues between a patient and a counselor conditioned on the provided inputs and following a set of conversational guidelines. Prompt details are provided in Appendix B.

Symptom-Only (SO). SO provides only a list of symptoms with their severity levels, with no demographic or situational context. This represents the highest privacy regime. We additionally consider variants that augment symptoms with synthetic profiles, with details provided in Appendix E.

Contextual Situation (CS). CS provides the patient's background situational description along with symptom information. While no direct quasi-identifiers are used, situational details may still indirectly leak personal information.

Full Profile (FP). FP further provides the generation model with patient demographics (e.g., age, gender, occupation, marital status) along with situation and a symptoms. This represents the lowest privacy regime, as generation is conditioned on rich personal information. Further details and examples are provided in Appendix A.

4 Evaluating Privacy–Utility Trade-offs

In this section, we outline the evaluation protocols used to assess both the utility and privacy of the generated synthetic counseling sessions.

4.1 Utility Evaluation

We evaluate utility through counseling quality and diversity in the generated synthetic sessions.

Counseling Quality. To measure counseling quality, we follow prior work (Lee et al., 2024; Qiu and Lan, 2026; Kim et al., 2025) and adopt an LLM-as-a-judge framework to evaluate generated sessions using two established psychological scales. The Cognitive Therapy Rating Scale (CTRS) (Aarons et al., 2012) measures counselor competence across six dimensions, covering general counseling skills (Understanding, Interpersonal Effectiveness, Collaboration) and CBT-specific skills (Guided Discovery, Focus, Strategy), with each dimension scored from 0–6. The Working Alliance Inventory (WAI) (Horvath and Greenberg, 1989; Bayerl et al., 2022) evaluates therapeutic alliance through Task, Goal, and Bond components, each rated on a 1–7 scale. Higher scores indicate stronger counseling quality. More details about the scales are provided in Appendix C.

Diversity. We use multiple metrics to capture lexical and situational diversity. Distinct-n (Qiu et al., 2024) measures the proportion of unique n-grams, while Expected Adjusted Distinct (EAD) (Liu et al., 2022) mitigates Distinct-n sensitivity to sequence length. Lexical Diversity Density (LDD) (Qiu and Lan, 2025) captures vocabulary richness and distribution across dialogues, and Ent-n (Zhang et al., 2018) measures the entropy of n-gram distributions, with higher values indicating greater diversity. For situational diversity, we extract patient situations using an LLM, encode them with a Sentence Transformer (Reimers and Gurevych, 2019), and compute average pairwise cosine similarity (SitSim), where lower similarity indicates greater diversity. Further details are provided in Appendix D.

4.2 Privacy Evaluation

We adopt a data-holder perspective and measure disclosure risk (Zhang et al., 2022) in generated sessions, assuming an attacker with access to source profiles. Under this threat model, motivated by the success of LLM-based attacks (Lermen et al., 2026), we evaluate privacy using LLM-based attribute extraction, membership inference, and linkage attacks. While stronger attacks may exist and could yield greater leakage, these methods provide a practical lower bound for such attackers, as they require minimal expertise, can be deployed off-the-shelf, and are scalable (Lermen et al., 2026).

Attribute Extraction Attack. We prompt an LLM to extract demographic attributes (name, age, gender, marital status, occupation) and patient situations from synthetic sessions. Prior work shows that combinations of such attributes can enable re-identification, either manually (Sweeney, 2002) or automatically (Narayanan and Shmatikov, 2008; Lermen et al., 2026; Staab et al., 2024b). The model selects values from a predefined attribute set and outputs *Cannot be Identified* when uncertain. Prompt details are provided in Appendix F.

We treat each attribute-value pair as a binary prediction problem against ground-truth patient attributes and define: True Positive as correctly recovered present attribute value; False Positive as incorrectly recovered value (wrong value for a present attribute or extraction of an absent attribute); True Negative as correctly unrecovered absent attribute; and False Negative as present attribute value returned as *Cannot be Identified*. Based on these definitions, we report True Positive Rate (TPR), False Positive Rate (FPR), True Negative Rate (TNR), False Negative Rate (FNR), Accuracy, Precision, Recall, and F1, as well as attribute leakage rate (the fraction of correctly recovered attributes among all present attributes). Situation leakage is measured using cosine similarity between sentence embeddings of extracted and source situations.

Membership Inference Attack (MIA). Membership inference attacks (MIA) determine whether a data point was involved in the generation process, providing a fundamental measure of privacy leakage. In our setting, MIA predicts whether a patient profile was used to generate any session. We construct three sets: (i) a *reference set* of profiles extracted from generated sessions (reused from the extraction attack), (ii) a *member set* of source profiles used for generation, and (iii) an equally sized *non-member set* of randomly sampled unused profiles. This forms a balanced binary classification task, where membership is inferred from similarity to reference profiles. Following prior work (Lermen et al., 2026), we serialize profile attributes and situations into text and encode them using the Qwen3-8B embedding model (Zhang et al., 2025), and classify profile x using their maximum cosine similarity to the reference set as the classification score:

$$score(x) = \max_{x_r \in R} CosSim(x, x_r), \quad (1)$$

where R denotes reference profile embeddings. We report AUC-ROC as a summary of attack performance across all operating points. However, such aggregate metrics can obscure whether an attack can confidently identify true members, as they average over all thresholds (Carlini et al., 2022). This high-confidence regime is critical in our setting: reliably inferring the membership of even a small number of real patients constitutes a serious privacy breach, and targeted re-identification from auxiliary datasets is a well-documented threat (Narayanan and Shmatikov, 2008; Lermen et al., 2026). We therefore also report TPR at low FPR thresholds (FPR $\leq$ 0.1, 0.05, and 0.01) (Carlini et al., 2022; Lermen et al., 2026), which captures how many members an attacker can recover while incurring minimal false positives.

Linkage Attack. Linkage attacks evaluate whether synthetic data can be connected back to the individuals from whom it was derived, a key criterion for anonymization under privacy regulations (GDPR, 2016). In our setting, the attack aims to match a synthetic counseling session to its source patient profile. We extract attributes and situations from synthetic sessions and compare them against all source profiles, framing linkage as a similarity-based binary classification task: given a pair of profiles, determine whether they originate from the same patient. We embed all profiles using the Qwen3-8B embedding model and linkage score is computed using cosine similarity:

$$score(x_i, x_j) = CosSim(x_i, x_j), \quad (2)$$

where x_i and x_j denote profile embeddings. Similar to MIA, we report the AUC-ROC along with TPR at low FPRs (FPR ≤ 0.1, 0.05, and 0.01).

5 Experimental Setup

Dataset. We conduct experiments on the Eeyore dataset (Liu et al., 2025a), which aggregates 3,805 counseling transcripts from four existing sources (Welivita et al., 2023; Malhotra et al., 2022; Liu et al., 2021; Wu et al., 2022). The dataset provides patient symptoms, situational descriptions, and demographic attributes (Name, Age, Gender, Occupation, and Marital Status) extracted from original sessions using GPT-4o (OpenAI, 2024). We select the first 500 profiles with non-empty demographic information and no duplicates, resulting in 310 unique patient profiles used as seeds for synthetic session generation. We select profiles with non-empty demographic information to ensure that the three privacy levels correspond to progressively richer input information, rather than overlapping settings where situation-only and situation-plus-demographics provide identical inputs. This subset also balances evaluation coverage and computational cost, while leaving unused profiles available for constructing the non-member set in MIA.

LLMs. We use Llama3.3-70B-Instruct (Meta, 2024) for synthetic session generation with temperature $T = 0.7$. For LLM-as-a-judge scoring and attribute/situation extraction, we use Qwen2.5-72B-Instruct (Yang et al., 2024) with $T = 0$ to ensure deterministic outputs. We additionally evaluate attack robustness across model scales using Qwen2.5-7B-Instruct, Qwen2.5-32B-Instruct (Yang et al., 2024), and Qwen3-32B-Instruct (Team, 2025) in thinking mode. Results show that attack performance is largely consistent across model scales for MIA and linkage attacks, while attribute extraction is more sensitive to scale, with Qwen2.5-7B exhibiting weaker performance. Further details on this are provided in Appendix I. We also swap the generator and attacker models in Appendix H (Qwen2.5-72B-Instruct for generation and Llama3.3-70B-Instruct for attacks) and observe consistent trends. Additional implementation details, including libraries and infrastructure, are provided in Appendix J.

6 Privacy-Utility Trade-offs

Key Findings

Providing richer private information **increases diversity** but does **not improve psychological quality**. However, it introduces **privacy risks**, making sessions more vulnerable to attribute extraction, linkage, and membership inference attacks.

Input	CTRS						WAI		
	General			CBT					
	U	I	C	D	F	S	Task	Goal	Bond
SO	4.42	5.85	4.11	4.01	4.00	3.99	5.82	5.59	5.82
CS	4.47	5.89	4.17	4.00	4.00	3.99	5.91	5.62	5.86
FP	4.36	5.84	4.15	4.00	4.00	3.99	5.90	5.63	5.82

Table 1: Counseling quality measured using CTRS and WAI. For CTRS: **U** (Understanding), **I** (Interpersonal Effectiveness), **C** (Collaboration), **D** (Guided Discovery), **F** (Focus), **S** (Strategy).

Input	Distinct				Lexical Diversity		Entropy				SitSim ↓
	D-1 ↑	D-2 ↑	D-3 ↑	EAD ↑	LDD-P ↑	LDD-C ↑	E-1 ↑	E-2 ↑	E-3 ↑	E-4 ↑	
SO	0.014	0.103	0.257	0.032	74.08	73.50	5.69	8.77	10.25	11.06	0.675
CS	0.016	0.126	0.314	0.038	91.49	85.76	5.78	9.00	10.55	11.36	0.533
FP	**0.017**	0.126	0.312	**0.039**	89.91	87.55	5.77	8.98	10.53	11.34	**0.522**
FP + Defense											
NER	**0.017**	0.129	0.319	0.038	91.18	**88.63**	5.78	8.98	10.54	11.35	0.533
CoarsProf	**0.017**	0.127	0.316	0.038	91.46	88.11	5.78	8.99	10.56	11.36	0.546
NoiseProf ($p = 0.2$)	**0.017**	**0.130**	0.322	**0.039**	91.97	86.76	5.78	8.99	10.56	11.36	0.528
NoiseProf ($p = 0.5$)	**0.017**	0.127	0.315	0.038	**92.98**	84.67	5.78	8.98	10.53	11.34	0.528
NoiseProf ($p = 0.8$)	**0.017**	**0.130**	**0.323**	**0.039**	92.74	87.07	**5.79**	**9.01**	**10.57**	**11.37**	0.532
NoiseIterMix ($p = 0.8$)	0.016	0.120	0.300	0.037	88.24	82.78	5.76	8.94	10.47	11.29	0.561

Table 2: Diversity evaluation results. **D-1/2/3**: Distinct-1/2/3; **EAD**: Expected Adjusted Distinct; **LDD-P/C**: Lexical Diversity Density (Patient/Counselor); **E-1–4**: Entropy at n-gram levels 1–4; **SitSim**: average cosine similarity across situations (lower = more diverse). **Bold** indicates best per column.

We analyze the privacy–utility trade-off in synthetic counseling session generation pipeline by varying the amount of private information provided as input: SO (symptoms), CS (symptoms + situation), and FP (symptoms + situation + demographics).

Utility Evaluation. We first evaluate counseling quality across privacy levels. As shown in Table 1, FP, CS, and SO achieve comparable CTRS and WAI scores, indicating that reducing patient information has limited impact on counseling quality. This suggests that the model's embedded psychological knowledge primarily drives counselor competence and therapeutic alliance, while additional patient private information provides limited gains.

Diversity evaluation results in Table 2 shows that FP and CS consistently improve diversity over SO across all metrics, demonstrating that demographic and situational information helps prevent repetitive narratives. However, demographics provide limited additional diversity beyond situations, suggesting that situational information is the main driver of diversity. Results for SO augmented with synthetic profiles (Appendix Table 8) show comparable or lower diversity than SO, suggesting that generations without grounding in real patient profiles produce less specific and more repetitive narratives, consistent with prior findings (Zhang, 2025). This has implications for downstream use of generated data, as diverse training data is important for improving counseling models (Chen et al., 2024).

Privacy Evaluation. Attribute extraction results (Table 3) show minimal leakage for SO due to the lack of patient private information in inputs, whereas CS and FP show substantial leakage. FP enables extraction of 59.19% of demographic attributes and extraction of situations highly similar to the original situations. CS also allows 46.62% attribute extraction despite lacking explicit demographics in the input, showing that situational descriptions can implicitly reveal demographic information. High TPR, low FPR, and strong accuracy further demonstrate that these attacks are both effective and reliable. Attribute-wise extraction results are provided in Appendix F.

Membership inference results (Table 4) show a similar trend. SO achieves near-random attack performance (0.539 AUC), and achieve TPR almost equal to the FPRs at the low FPR thresholds, indicating an unreliable inference. In contrast, MIA on CS and FP achieves AUC above 0.75 with high TPRs at low FPR thresholds, indicating substantial membership leakage.

Input	Confusion Metrics				Classification Metrics				Leakage	Similarity
	TPR ↓	FPR ↑	TNR ↓	FNR ↑	Acc ↓	Prec ↓	Recall ↓	F1 ↓	Attr (%) ↓	CosSim ↓
SO	0.073	0.122	0.878	0.927	0.586	0.253	0.073	0.113	6.43	0.581 ± 0.13
CS	0.584	0.283	0.717	0.416	0.674	0.501	0.584	0.539	46.62	0.732 ± 0.10
FP	0.728	0.294	0.706	0.272	0.714	0.552	0.728	0.628	59.19	0.722 ± 0.11
FP + Defense										
NER	0.582	0.283	0.717	0.418	0.673	0.501	0.582	0.538	46.57	0.720 ± 0.10
CoarsProf	**0.427**	0.261	0.739	**0.573**	0.639	0.438	**0.427**	0.433	33.61	0.706 ± 0.10
NoiseProf ($p = 0.2$)	0.645	0.360	0.640	0.355	0.641	0.426	0.645	0.513	46.52	0.716 ± 0.10
NoiseProf ($p = 0.5$)	0.546	0.444	0.556	0.454	0.553	0.302	0.546	0.388	34.73	0.697 ± 0.11
NoiseProf ($p = 0.8$)	0.461	0.518	0.482	0.539	0.477	0.206	0.461	0.284	25.40	0.692 ± 0.11
NoiseIterMix ($p = 0.8$)	0.460	**0.532**	**0.468**	0.540	**0.466**	**0.189**	0.460	**0.268**	**23.87**	$\mathbf{0.638 \pm 0.11}$

Table 3: Attribute extraction attack performance. **TPR/FPR/TNR/FNR**: True/False Positive/Negative Rates; **Acc**: Accuracy; **Prec**: Precision; **Attr (%)**: percentage of attribute leakage; **CosSim**: cosine similarity between original and extracted situations. **Bold** indicates the best defense per column.

Input	AUC ↓	TPR @ FPR $\leq \tau$ ↓		
		τ=0.10	τ=0.05	τ=0.01
SO	0.539	0.109	0.043	0.010
CS	0.769	0.490	0.405	0.194
FP	0.811	0.492	0.412	0.233
FP + Defense				
NER	0.743	0.378	0.284	0.134
CoarsProf	0.715	0.331	0.238	0.134
NoiseProf ($p = 0.2$)	0.748	0.441	0.342	0.211
NoiseProf ($p = 0.5$)	0.712	0.327	0.224	0.135
NoiseProf ($p = 0.8$)	0.700	0.316	0.194	0.072
NoiseIterMix ($p = 0.8$)	**0.631**	**0.185**	**0.116**	**0.053**

Table 4: Membership inference attack performance. **AUC**: area under the ROC curve; **TPR @ FPR $\leq$ threshold** (τ) : attack performance at controlled false positive rates. **Bold** indicates the strongest defense per column.

Input	AUC ↓	TPR @ FPR $\leq \tau$ ↓		
		τ=0.10	τ=0.05	τ=0.01
SO	0.608	0.188	0.102	0.026
CS	0.968	0.928	0.878	0.734
FP	0.985	0.970	0.920	0.784
FP + Defense				
NER	0.970	0.926	0.866	0.739
CoarsProf	0.961	0.903	0.829	0.639
NoiseProf ($p = 0.2$)	0.955	0.878	0.813	0.586
NoiseProf ($p = 0.5$)	0.924	0.799	0.696	0.508
NoiseProf ($p = 0.8$)	0.896	0.730	0.632	0.421
NoiseIterMix ($p = 0.8$)	**0.794**	**0.541**	**0.416**	**0.198**

Table 5: Linkage attack performance. **AUC**: area under the ROC curve; **TPR @ FPR $\leq$ threshold** (τ): attack performance at controlled false positive rates. **Bold** indicates the strongest defense per column.

Finally, linkage attacks (Table 5) reveal that SO provides limited linkage risk, while CS and FP remain highly linkable, achieving AUC over 0.95 and TPR above 0.9 at FPR of 0.1. Even at stricter thresholds (FPR $\leq$ 0.01), TPR remains around 0.75, demonstrating that additional patient private information significantly increases the risk of linking generated sessions to source profiles.

7 Defenses

We investigate four defenses to improve privacy while preserving the benefits of incorporating patient private information during generation. These include one post-hoc defense applied to generated sessions and three pre-generation defenses that modify input profiles before generation. Additional details, including prompts, perturbation statistics, and examples are provided in Appendix G.

NER-based De-identification (NER). We use an LLM-based NER pipeline to identify sensitive attributes (e.g., names, age, occupation, marital status, organizations, gender) and replace them with categorical placeholders (e.g., [NAME]), removing explicit identifiers while preserving conversational content. Unlike encoder-based approaches, this pipeline requires no task-specific annotations and can flexibly adapt to new attribute categories via prompting, while achieving performance comparable to or exceeding fine-tuned clinical NER models in low-resource settings (Liu et al., 2023b).

Coarsened Profile (CoarsProf). This defense reduces exposure by coarsening profile attributes. Names are removed, occupation and marital status are mapped to coarse categories, and age and gender are retained as categorical variables. The patient situation is adapted to the coarsened profile using an LLM to remove or generalize details while preserving clinical and interpersonal context.

Noisy Profile (NoiseProf). This defense introduces randomized profile perturbations into the source profile. Each attribute is independently replaced with probability p by uniformly sampling from the corresponding attribute pool collected from source profiles; missing attributes are added with the same probability. Names are removed and symptoms are kept unchanged. The situation is then adapted using an LLM to maintain coherence with the perturbed profile. We evaluate $p \in \{0.2, 0.5, 0.8\}$.

Noisy Profile Iterative Mixing (NoiseIterMix). Building on NoiseProf ($p = 0.8$), this method further diversifies the situational description through iterative mixing. At each iteration, a situation from another profile sharing a demographic attribute is retrieved, relevant aspects are incorporated into the target situation, and the result is rewritten to remain consistent with the final profile. The process runs for up to five iterations or until cosine similarity with the original situation falls below 0.6.

8 Impact of Defenses

Key Findings

All defenses **preserve diversity** comparable to FP, but surface-level methods (NER and CoarsProf) offer limited privacy gains due to indirect leakage, whereas **noise-based approaches** (NoiseProf and NoiseIterMix) **are more effective**.

In this section, we present defense results on FP generations, while results for defenses on CS are provided in Appendix G. Defenses on CS maintain comparable diversity and provide slight improvements against attribute extraction attacks, as CS generations do not explicitly use demographic information. However, while achieving comparable MIA protection to defenses on FP, they offer weaker linkage protection, as perturbing only situations causes less divergence from original sessions than jointly perturbing demographic attributes and situations. Table 2 shows that defenses largely preserve diversity, with only a slight decrease for NoiseIterMix ($p = 0.8$), indicating limited utility degradation after defense modifications.

For attribute extraction attacks (Table 3), NER removes explicit quasi-identifiers but remains vulnerable to indirect leakage, resulting in leakage comparable to CS. CoarsProf reduces attribute leakage by $\sim 25\%$ relative to FP, while NoiseProf provides increasing protection with higher perturbation probability p, achieving a $\sim 35\%$ reduction at $p = 0.8$. Unlike CoarsProf, which mainly lowers TPR, NoiseProf also increases FPR, making extraction less reliable. NoiseIterMix ($p = 0.8$) provides comparable attribute protection to NoiseProf ($p = 0.8$) while further reducing situational leakage measured by cosine similarity of the extracted situation to the original situation.

For MIA (Table 4), NER and CoarsProf remain vulnerable, achieving AUC over 0.7. While they help to reduce TPR at low FPR thresholds, TPR still remains significantly higher than FPRs around 3-10 times, indicating that generated sessions still retain identifiable signals from members. NoiseProf provides increasing protection as the perturbation probability p increases, however even $p = 0.8$, it provides similar protection to CoarsProf and NER. NoiseIterMix ($p = 0.8$) achieves stronger protection, lowering AUC to around 0.63 and substantially reducing TPR under low-FPR thresholds, showing the importance of perturbing the demographics as well as diverging the situation.

Linkage attack results (Table 5) follow similar trends. NER and CoarsProf provide limited protection and remain highly vulnerable. NoiseProf improves protection as perturbation probability p increases, with NoiseProf ($p = 0.8$) reducing AUC by $\sim 7\%$, although linkage remains effective due to preserved situational similarity. NoiseIterMix ($p = 0.8$) further reduces AUC by $\sim 10\%$ over NoiseProf ($p = 0.8$) and substantially lowers TPR under low-FPR thresholds, demonstrating the importance of perturbing attributes and diverging situations. Overall, effective protection requires modifying explicit attributes and situational information, as surface-level de-identification is insufficient.

9 Conclusion

In this work, we systematically study the privacy–utility trade-off in synthetic counseling session generation under varying levels of private input information. Our results show that incorporating patient private information improves diversity. However, these gains introduce privacy risks, with richer inputs increasing vulnerability to attribute extraction, membership inference, and linkage attacks. We further evaluate defenses on Full Profile (FP) generations and find that approaches, such as NER-based redaction and profile coarsening, provide limited protection. In contrast, our proposed noise-based method, NoiseIterMix, substantially reduces linkage and membership inference attack effectiveness by 20% each and attribute extraction leakage by 35%, while preserving diversity.

Limitations

While this work provides a systematic evaluation of privacy-utility trade-offs in synthetic counseling session generation and investigates the effectiveness of some defenses, several limitations remain.

Our utility evaluation relies on LLM-as-a-judge assessments to measure the psychological quality of generated counseling sessions. Recent studies have demonstrated that LLM-based evaluations on psychological criteria achieves strong correlation with human expert judgments (Lee et al., 2024; Chen et al., 2025). However, concerns remain regarding the reliability, validity, and potential biases of using psychological scales like CTRS and WAI as a measure of psychological quality.

Our privacy evaluation considers an auditing scenario in which the evaluator has access to the original source data and uses LLM-based attacks to assess potential privacy leakage. This setting reflects a data-holder perspective, where the goal is to identify whether sensitive information from the source data may unintentionally appear in released synthetic sessions. Under this threat model, our use of simple, off-the-shelf LLM-based attacks provides a practical lower bound on leakage for such attackers, while more sophisticated or tailored attacks may reveal higher risks. However, this auditing setup represents a simplified threat model, as realistic attackers may have limited or no access to the original data. Future work should investigate privacy risks under more constrained threat models, including settings with auxiliary information and black-box access. Additionally, while LLM-based attacks provide a flexible mechanism for evaluating semantic privacy leakage, their effectiveness can be influenced by factors such as prompt design and the choice of attack model. Exploring more deterministic and model-agnostic privacy attacks would provide a more comprehensive understanding of privacy risks in synthetic counseling data generation.

Finally, the defense strategies explored in this work should be viewed as preliminary approaches for reducing private information leakage while preserving the utility of generated sessions. Although these methods demonstrate reduced vulnerability against the evaluated LLM-based attacks, they do not provide formal privacy guarantees. Future research should investigate privacy-preserving synthetic counseling generation frameworks that provide formal privacy guarantees while maintaining the counseling quality and diversity of generated data.

Ethics

This work investigates the privacy-utility trade-off in synthetic counseling session generation pipelines by systematically evaluating how varying amounts of private input information affect the utility and privacy risks of generated sessions. To assess potential privacy leakage, we introduce LLM-based privacy attacks, including attribute extraction attacks, membership inference attacks, and linkage attacks. While these attacks demonstrate possible privacy risks associated with synthetic counseling data generation, we emphasize that they are designed from the perspective of a data holder performing privacy auditing and assume that the attacker already has access to the original source data. Such an assumption is considerably stronger than realistic adversarial settings, particularly for sensitive counseling data that are typically not publicly released due to ethical and privacy considerations.

We acknowledge that publishing attack methodologies may potentially enable the development of more advanced privacy attacks. However, we believe that the risks associated with this work are limited and are outweighed by its potential benefits. In particular, the proposed privacy evaluation framework provides synthetic counseling session generators with tools to proactively assess privacy risks before releasing synthetic counseling datasets. Such auditing is essential because synthetic data generation does not inherently guarantee privacy preservation, and generated samples may unintentionally retain or reveal sensitive information from the underlying data. By encouraging systematic privacy evaluation, this work aims to promote more responsible development and deployment of synthetic counseling resources.

This study does not release any new dataset. All experiments are conducted using the publicly available Eeyore dataset (Liu et al., 2025a), and we use publicly available open-source language models from the Llama and Qwen model families. We hope that this work motivates future research on synthetic counseling session generation to incorporate privacy auditing as a standard practice, ensuring that generated datasets provide meaningful utility while minimizing unintended disclosure of sensitive information.

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A Profile details

In this section, we present detailed statistics on the distribution of demographic and clinical attributes in the dataset, along with a representative example of a patient profile. As described in Section 5, we filter the data to retain only unique profiles that contain at least one demographic attribute, considering the first 500 data points. This selection ensures that all three privacy levels of input are associated with distinct inputs, with each level corresponding to progressively richer amounts of private information. This results in a total of 310 profiles used in our analysis.

Each profile comprises a subset of demographic attributes, including Name, Gender, Age, Marital Status, and Occupation. In addition, profiles include a presenting situation, which describes the patient's current concerns and contextual circumstances. The clinical information in the profiles capture various mental health indicators, such as depression severity, levels of suicidal and homicidal ideation, the severity of different psychological symptoms, and the presence of cognitive distortions. To further characterize the patient's disposition, each profile also specifies the degree of resistance toward counseling, which helps inform the interaction dynamics. While some profiles additionally include prior counseling history, we do not utilize this information during generation. Instead, all synthetic dialogues are constructed under the assumption that they represent an initial counseling session.

We report the distributions of key demographic attributes: Age, Gender, Marital Status, and Occupation, as well as patient resistance levels across the 310 profiles in Table 6. The distributions of clinical attributes, including cognitive distortions, symptom severity, suicidal ideation, homicidal ideation, and depression severity, are summarized in Table 7. Finally, a representative example of a patient profile is illustrated in Figure 2.

B Synthetic Counseling Session Generation

As described in Section 3, we generate synthetic counseling sessions under three privacy levels of input information, where each setting provides the LLM with a different subset of patient background information: (i) symptoms only, (ii) symptoms with contextual situation information, and (iii) symptoms with contextual situation and demographic information. These settings are designed to simulate varying degrees of private information disclosure during synthetic session generation.

For all generation settings, the LLM receives a structured prompt consisting of global generation guidelines as well as patient- and counselor-turn specific instructions. The provided patient information is organized into two broad categories: (1) **background information**, which includes demographic attributes and details about the patient's presenting situation, and (2) **clinical information**, which includes symptom severity, cognitive distortions, and indicators of suicidal and homicidal ideation. This separation allows us to systematically control the amount of private background information revealed to the model while keeping the clinical context consistent across different privacy settings.

The three generation settings differ in the extent of background information provided to the model. In the **Symptoms-Only (SO)** setting, the model receives only clinical information along with the patient's resistance level, without any additional contextual or demographic details. The prompt used for SO generation is provided in Figure 3.

In the **Contextual Situation (CS)** setting, the model is provided with additional background information in the form of the patient's presenting situation, while demographic attributes remain undisclosed. The prompt template used for CS generation is provided in Figure 4.

Finally, in the **Full Profile (FP)** setting, the model receives the complete patient background information, including both contextual situation and demographic attributes, in addition to the clinical information. This setting represents the highest level of private information disclosure among the three configurations. The prompt template used for FP generation is provided in Figure 5.

C Psychological Scales

To assess the utility of the generated synthetic sessions, we evaluate counseling quality as one of the aspects. For evaluating counseling quality, following prior works (Lee et al., 2024; Qiu and Lan, 2026; Kim et al., 2025), we use LLM-as-a-judge evaluation on two psychological scales: Cognitive Therapy Rating Scale (CTRS) (Aarons et al., 2012) and the Working Alliance Inventory (WAI) (Horvath and Greenberg, 1989).

The CTRS is used to assess both general counseling competencies and skills specific to cognitive behavioral therapy (CBT).

General counseling abilities are evaluated along the following dimensions:

- **Understanding:** The extent to which the

Age Group	Ct.
0~24	145
25~44	104
45~64	6
65+	2
N/A	53

(a) Age

Gender	Ct.
Male	85
Female	68
N/A	157

(b) Gender

Resistance	Ct.
Low	192
Medium	100
High	18

(c) Resistance

Marital Status	Ct.
Married	36
Single	23
single	2
In a relationship	8
In a relationship, unmarried	1
Not married	1
Unmarried	1
Divorced	3
Separated	2
Breakup with husband	1
Widowed	1
widowed	1
Engaged	1
N/A	229

(d) Marital status

Occupation	Count	Occupation	Count
Student	56	Hotel or Conference Center Worker	1
College student	4	Security worker	1
PhD student	1	Cleaning job	1
Film Student and Podcast Host	1	Intern	1
Medical student	1	Part-time job	2
Graduate (master's field)	1	Commission based job	1
Academic Research Associate	1	Self employed	1
Recent college grad. (econ. research)	1	Works from home	1
Doctor	2	Accounting (bookkeeper)	1
Retired Nurse	1	Peace Corps volunteer	1
Computer Science Engineer	1	Military	1
Electrical engineer	1	Computer stuff	1
Engineer in Mechanical Engineering	1	PhD in immunology	1
Technical support specialist	1	Unemployed	1
Customer service and technical expert	1	Hair braider	1
Tech person	1	Special education teacher	1
McDonald's employee	2	Asst. relig. ed. teacher/daycare	1
Home Depot employee	2	Employee (managerial/admin.)	1
Kmart employee	1	Non-profit library system employee	1
Retail	1	Music industry professional	1
Gas station worker	1	Drummer	1
Teller at a bank	1	Writer	1
Bank employee	1	N/A	201
Call center employee	1		

(e) Occupation

Table 6: Distribution of demographic attributes (Age, Gender, Marital Status, Occupation) and patient resistance levels across the 310 selected profiles. **Ct.**: Count.

Cognitive Distortion	Ct.
Overgeneralizing	180
Selective Abstraction	159
Catastrophic Thinking	153
Minimisation	128
Personalization	115
Arbitrary Inference	62

(a) Cognitive distortion

Suicidal Ideation	Ct.
No	207
Mild	18
Moderate	51
Severe	33
N/A	1

(b) Suicidal ideation

Homicidal Ideation	Ct.
No	302
Mild	4
Moderate	2
Severe	1
N/A	1

(c) Homicidal ideation

Depression Severity	Ct.
Minimal	44
Mild	82
Moderate	113
Severe	71

(d) Depression severity

Symptom	Count
Feelings of sadness, tearfulness, emptiness, or hopelessness	284
Anxiety, agitation, or restlessness	274
Becoming withdrawn, negative, or detached	233
Feelings of worthlessness or guilt, fixating on past failures or self-blame	232
Loss of interest or pleasure in most or all normal activities, such as sex, hobbies, or sports	207
Trouble thinking, concentrating, making decisions, and remembering things	204
Tiredness and lack of energy, so even small tasks take extra effort	162
Angry outbursts, irritability, or frustration, even over small matters	156
Frequent or recurrent thoughts of death, suicidal thoughts, suicide attempts, or suicide	106
Sleep disturbances, including insomnia or sleeping too much	99
Greater impulsivity	83
Increased use of alcohol or drugs	69
Increased engagement in high-risk activities	66

(e) Depressive symptoms

Table 7: Distribution of clinical attributes, including cognitive distortions, depression severity, levels of suicidal and homicidal ideation, and symptoms across the 310 selected profiles. **Ct.**: Count.

counselor accurately comprehends and interprets the patient's concerns and problems.

- **Interpersonal Effectiveness:** The counselor's capacity to establish and sustain a supportive and effective therapeutic relationship.
- **Collaboration:** The degree to which the counselor actively engages the patient in shared goal-setting and decision-making processes.

CBT-specific competencies are assessed through the following aspects:

- **Guided Discovery:** The effectiveness with which the counselor facilitates patient insight through structured questioning and reflective dialogue.
- **Focus:** The counselor's ability to identify and concentrate on the most relevant thoughts and behaviors for change.
- **Strategy:** The appropriateness and coherence of the therapeutic approach used to promote cognitive and behavioral change.

Each dimension is scored on a scale from 0 to 6, where higher values reflect greater proficiency in the corresponding skill. The evaluation prompt used in this study is illustrated in Figure 6.

The Working Alliance Inventory (WAI) focuses on assessing the quality of the therapeutic relationship between the counselor and the patient. To compute WAI scores, we adopt the evaluation protocol introduced by Qiu and Lan (2026). The inventory consists of 12 items organized into three primary dimensions: Task, Goal, and Bond. The Task dimension reflects the patient's understanding of and agreement with the therapeutic activities; Goal captures the level of alignment between counselor and patient regarding the objectives of therapy; and Bond represents the strength of the emotional connection and mutual trust within the relationship.

The 12 WAI items and their corresponding dimensions are as follows:

- **WAI-1 (Task):** The patient and counselor are aligned on the steps being taken to improve the patient's situation.
- **WAI-2 (Task):** The patient and counselor agree on the value in the ongoing counseling process and the patient gains new insights into their concerns.
- **WAI-3 (Bond):** The patient and counselor share a sense of mutual liking.
- **WAI-4 (Goal):** There is uncertainty or lack of clarity regarding the objectives of counseling.

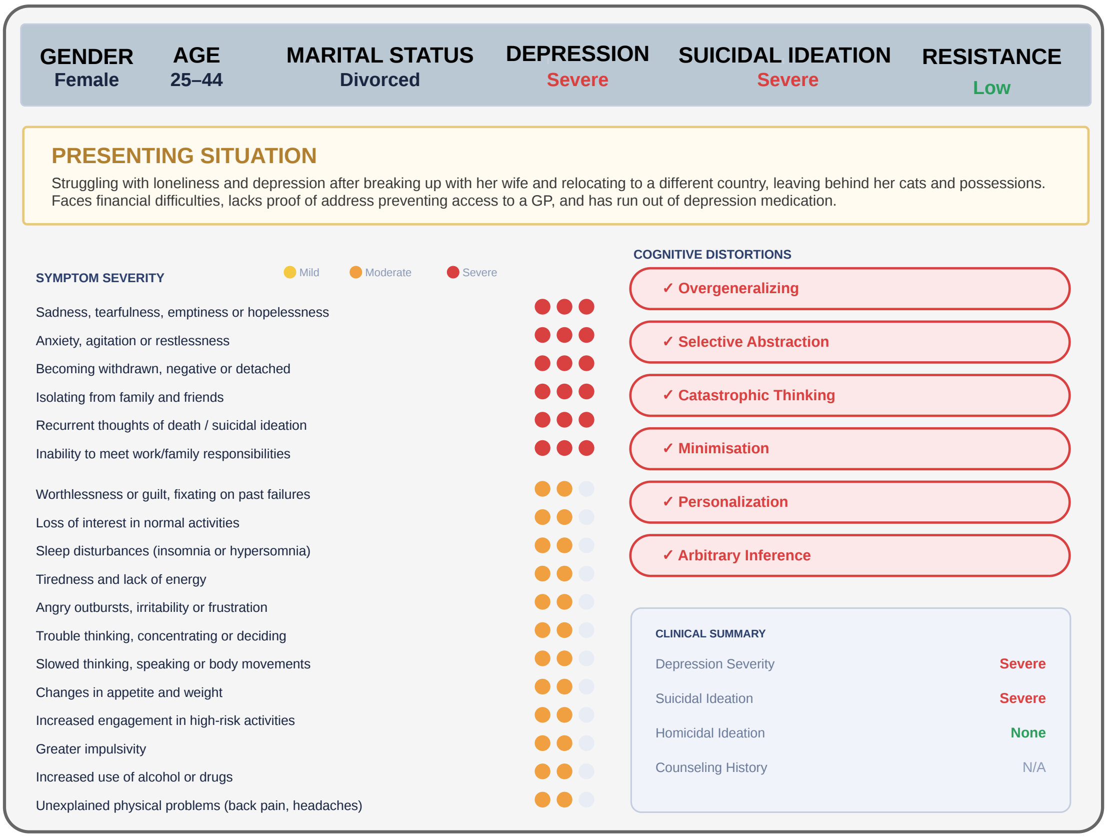

Figure 2: Representative example of a patient profile illustrating demographic attributes, presenting situation, patient resistance and clinical attributes used for synthetic counseling session generation.

- **WAI-5 (Bond):** The patient trusts the counselor's ability to provide meaningful support.
- **WAI-6 (Goal):** The patient and counselor work together toward mutually agreed-upon goals.
- **WAI-7 (Bond):** The patient feels respected and valued by the counselor.
- **WAI-8 (Task):** Both parties agree on the key issues that need to be addressed.
- **WAI-9 (Bond):** A mutual trust exists between the patient and counselor.
- **WAI-10 (Goal):** The patient and counselor hold differing views about the patient's main concerns.
- **WAI-11 (Goal):** There is shared clarity about the changes that would benefit the patient.
- **WAI-12 (Task):** The patient perceives the therapeutic approach as appropriate and effective.

Each item is rated on a 1–7 scale, where higher scores indicate a stronger therapeutic alliance except for negatively phrased items (WAI-4 and WAI-10). For these items, lower scores correspond to a better alliance and therefore, these items are reverse-scored by subtracting their values from 8 before aggregation. The final scores for each dimension are computed as follows:

$$\begin{aligned} Score_{Task} = (&Score_{wai-1} + Score_{wai-2} \\ &+ Score_{wai-8} + Score_{wai-12})/4 \end{aligned}$$

$$\begin{aligned} Score_{Goal} = ((&8 - Score_{wai-4}) + Score_{wai-6} \\ &+ (8 - Score_{wai-10}) + Score_{wai-11})/4 \end{aligned}$$

$$\begin{aligned} Score_{Bond} = (&Score_{wai-3} + Score_{wai-5} \\ &+ Score_{wai-7} + Score_{wai-9})/4 \end{aligned}$$

The prompt used in our WAI evaluation is shown in Figure 7.

D Diversity Evaluation

To quantify the diversity of the generated counseling sessions, we use the Distinct-n metrics

($n \in \{1, 2, 3\}$) (Qiu et al., 2024), which measure the proportion of unique n-grams among all n-grams in the generated corpus. Larger Distinct-n values reflect higher lexical diversity. Before computing these metrics, we merge all counselor and patient utterances within each generated dialogue, remove punctuation, and tokenize the resulting text using the NLTK tokenizer.

Since Distinct-n can be affected by sequence length, with longer texts often receiving lower scores due to increased repetition, we additionally report the Expectation-Adjusted Distinct (EAD) metric (Liu et al., 2022). EAD compensates for this length dependency by comparing the observed lexical diversity against the expected distinctness for a sequence of the same length. This adjustment makes EAD a more reliable measure of lexical variation and improves its alignment with human assessments of diversity. The EAD score is computed as:

$$EAD = \frac{N}{V[1 - (\frac{V-1}{V})^C]} \tag{3}$$

where N denotes the number of unique tokens, C represents the total number of tokens in the sequence, and V corresponds to the vocabulary size.

In addition to Distinct-n and EAD, we evaluate diversity using Lexical Diversity Density (LDD) (Qiu and Lan, 2025), computed separately for patient and counselor utterances. LDD measures the proportion of unique words relative to the total number of words while accounting for the number of generated dialogues. It captures not only vocabulary richness but also whether diverse lexical choices are consistently distributed across sessions. The metric is defined as:

$$LDD = \frac{\text{Total Unique Words}}{\text{Total Words} \times \text{\# of Dialogues}} \tag{4}$$

Higher LDD values indicate that the generated sessions contain a broader range of unique vocabulary across the evaluated dialogues.

We further assess lexical diversity using entropy-based measures, Ent-n (Zhang et al., 2018). Given the distribution of n-grams in the generated corpus, Ent-n measures the uncertainty of this distribution and is defined as:

$$\text{Ent-}n = -\sum_{w} p(w) \log(p(w)) \tag{5}$$

where $p(w)$ represents the probability of observing n-gram w within the complete corpus. A higher entropy value indicates a more uniform distribution of n-grams, suggesting greater variation in generated language.

Beyond lexical variation, we evaluate diversity in the underlying patient scenarios represented in generated sessions. To measure situational diversity, we first use an LLM to extract a concise summary of the patient's presenting situation from each generated dialogue. These extracted descriptions are then encoded into dense semantic representations using a Sentence Transformer model (Reimers and Gurevych, 2019). We compute the pairwise cosine similarity between all situation embeddings and report the average similarity across the generated sessions. Lower similarity scores indicate that generated dialogues cover a wider range of patient situations, while higher scores suggest greater overlap between scenarios.

Formally, given N generated sessions with corresponding situation embeddings $\mathbf{e}_1, \ldots, \mathbf{e}_N$, the average situational similarity is calculated as:

$$\text{SitSim} = \frac{2}{N(N-1)} \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} \frac{\mathbf{e}_i \cdot \mathbf{e}_j}{\|\mathbf{e}_i\| \, \|\mathbf{e}_j\|}, \tag{6}$$

where the cosine similarity term measures the semantic closeness between two extracted situations.

E Analysis of Session Diversity under Synthetic Profile Generation

To further analyze the diversity of sessions generated using synthetically constructed patient profiles, we introduce two additional baselines: (i) **Randomized Profile (RProf)** and (ii) **Symptom-based Profile (SProf)**. These baselines evaluate whether synthetic priors can achieve comparable diversity to profiles derived from real patient information while reducing reliance on private attributes.

(i) **Randomized Profile (RProf).** In this setting, we construct synthetic profiles by randomly sampling demographic attributes from the pool of possible attribute values observed across the real profiles. These sampled demographic attributes are then combined with the original clinical information (i.e., symptoms) from real profiles. Conditioned on this information, we prompt the LLM to generate a synthetic patient situation, completing the construction of a synthetic patient profile. The prompt used for generating the synthetic situation is provided in Figure 8. The generation used a temperature of $T = 0.7$. The resulting synthetic profiles are subsequently used as inputs for synthetic counseling session generation.

(ii) **Symptom-based Profile (SProf).** In this setting, we generate both demographic attributes and presenting situations synthetically, conditioned only on the clinical information (i.e., symptoms). The prompt used for jointly generating demographic attributes and situational information is provided in Figure 9. The generation used a temperature of $T = 0.7$. The resulting fully synthetic profiles are then used to generate synthetic counseling sessions.

We compare sessions generated using RProf and SProf with our three privacy-level generated sessions: Symptoms-Only (SO), Contextual Situation (CS), and Full Profile (FP), as well as the proposed defenses. The diversity results are presented in Table 8. We observe that both RProf and SProf achieve diversity levels comparable to or lower than the SO generations. This suggests that, in the absence of grounding from real patient profiles, LLMs tend to generate less varied and more generic background narratives, resulting in reduced situational specificity and diversity (Zhang, 2025). Furthermore, the defense mechanisms preserve diversity levels close to those achieved by FP, indicating that they provide a more favorable utility compared to relying on fully synthetic profile generation for counseling session synthesis.

F LLM-based Attribute Extraction

We use LLM-based attribute extraction to quantify the extent to which private patient demographic and situational information can be recovered from synthetic counseling sessions. For each demographic attribute, the LLM is prompted to infer the corresponding attribute value from a predefined set of candidate values. To encourage conservative predictions, the prompt explicitly instructs the model to output *Cannot be Identified* when the available evidence is insufficient to make a reliable inference.

We design separate extraction prompts for each demographic attribute, including name (Figure 10), age (Figure 12), gender (Figure 11), marital status (Figure 14), and occupation (Figure 13). For extracting the patient's background situation, we augment the prompt with two in-context examples to provide guidance on the expected level of detail, format, and content of the extracted information. The prompt used for situation extraction is provided in Figure 15.

To further analyze the results of the LLM-based attribute extraction attacks, we report attribute-wise extraction performance in Table 9. In the **SO** setting, where no demographic attributes are provided during generation, most attributes are not recoverable. Age shows a relatively higher extraction rate compared to other attributes, however, the high false positive rate indicates that these predictions are unreliable and do not represent meaningful privacy leakage. This suggests that synthetic sessions generated without demographic and background situation as input contain limited identifiable demographic information.

In contrast, the **CS** setting exhibits higher true positive rates across several attributes, particularly for age and marital status. This indicates that contextual situation information can implicitly encode demographic cues, even when explicit demographic attributes are withheld. For example, descriptions of academic stress, employment-related concerns, or career transitions may provide indirect signals about age, while discussions of interpersonal or family-related difficulties may reveal information related to marital status.

The **FP** setting demonstrates consistently higher leakage across all evaluated attributes. Since demographic information is explicitly provided as part of the generation input, the resulting sessions may preserve and reflect these attributes, making them more susceptible to attribute extraction attacks. These results highlight the privacy risks associated with incorporating detailed patient profiles into synthetic counseling session generation.

G Defense details

We evaluate four defense strategies to reduce privacy leakage in synthetic counseling sessions: (i) NER-based redaction of generated sessions, (ii) generation from coarsened patient profiles, (iii) generation from noisy profiles with perturbed demographic attributes, and (iv) generation from iteratively diversified noisy profiles through background situation modification.

In NER-based redaction, we apply LLM-based named entity recognition (NER) and redaction to remove explicit identifiers from generated counseling sessions. The LLM is instructed through its system prompt to replace personally identifiable information with predefined placeholders,

Input	Distinct				Lexical Diversity		Entropy				SitSim ↓
	D-1 ↑	D-2 ↑	D-3 ↑	EAD ↑	LDD-P ↑	LDD-C ↑	E-1 ↑	E-2 ↑	E-3 ↑	E-4 ↑	
SO	0.014	0.103	0.257	0.032	74.08	73.50	5.69	8.77	10.25	11.06	0.675
CS	0.016	0.126	0.314	0.038	91.49	85.76	5.78	9.00	10.55	11.36	0.533
FP	**0.017**	0.126	0.312	**0.039**	89.91	87.55	5.77	8.98	10.53	11.34	**0.522**
RProf	0.013	0.096	0.246	0.034	75.33	69.93	5.64	8.71	10.25	11.11	0.558
SProf	0.012	0.087	0.224	0.031	69.57	66.05	5.59	8.59	10.09	10.95	0.649
FP + Defense											
NER	**0.017**	0.129	0.319	0.038	91.18	**88.63**	5.78	8.98	10.54	11.35	0.533
CoarsProf	**0.017**	0.127	0.316	0.038	91.46	88.11	5.78	8.99	10.56	11.36	0.546
NoiseProf ($p = 0.2$)	**0.017**	**0.130**	0.322	**0.039**	91.97	86.76	5.78	8.99	10.56	11.36	0.528
NoiseProf ($p = 0.5$)	**0.017**	0.127	0.315	0.038	**92.98**	84.67	5.78	8.98	10.53	11.34	0.528
NoiseProf ($p = 0.8$)	**0.017**	**0.130**	**0.323**	**0.039**	92.74	87.07	**5.79**	**9.01**	**10.57**	**11.37**	0.532
NoiseIterMix ($p = 0.8$)	0.016	0.120	0.300	0.037	88.24	82.78	5.76	8.94	10.47	11.29	0.561

Table 8: Diversity evaluation for generations using synthetic priors (SProf and RProf). **D-1/2/3**: Distinct-1/2/3; **EAD**: Expected Adjusted Distinct; **LDD-P/C**: Lexical Diversity Density (Patient/Counselor); **E-1–4**: Entropy at n-gram levels 1-4; **SitSim**: average cosine similarity across situations (lower = more diverse). **Bold** indicates best per column.

Input	Attribute	TPR	FPR	TNR	FNR	Acc	Prec	Recall	F1	Attr (%)
SO	Name	0.000	0.000	1.000	1.000	0.898	Und.	0.000	Und.	0.00
	Gender	0.000	0.000	1.000	1.000	0.503	Und.	0.000	Und.	0.00
	Age	0.202	0.766	0.234	0.798	0.214	0.314	0.202	0.246	15.42
	Occupation	0.000	0.115	0.885	1.000	0.605	0.000	0.000	Und.	0.00
	Marital Status	0.013	0.040	0.960	0.988	0.711	0.100	0.013	0.022	1.23
CS	Name	0.200	0.004	0.996	0.800	0.918	0.857	0.200	0.324	19.35
	Gender	0.378	0.128	0.872	0.621	0.632	0.737	0.378	0.500	37.09
	Age	0.802	0.906	0.094	0.198	0.530	0.586	0.802	0.677	59.76
	Occupation	0.443	0.309	0.691	0.557	0.641	0.265	0.443	0.331	24.77
	Marital Status	0.718	0.373	0.627	0.282	0.648	0.370	0.718	0.488	63.75
FP	Name	0.963	0.018	0.982	0.037	0.980	0.839	0.963	0.897	81.25
	Gender	0.473	0.137	0.863	0.527	0.671	0.769	0.473	0.586	46.67
	Age	0.814	0.969	0.031	0.186	0.561	0.638	0.814	0.716	66.14
	Occupation	0.870	0.356	0.644	0.130	0.684	0.348	0.870	0.497	44.34
	Marital Status	0.821	0.372	0.628	0.179	0.671	0.387	0.821	0.526	72.37

Table 9: Attribute-wise leakage analysis under different inputs (SO, CS, FP). **TPR/FPR/TNR/FNR**: True/False Positive/Negative Rates; **Acc**: Accuracy; **Prec**: Precision; **Attr (%)**: percentage of attribute leakage; **Und.** indicates undefined values due to zero TPR.

including names with [NAME], ages with [AGE], occupations with [OCCUPATION], marital statuses with [MARITAL_STATUS], genders with [GENDER], organization names with [ORGANIZATION], and contact information (including phone numbers, email addresses, and physical addresses) with [CONTACT_INFO]. The prompt used for this defense is provided in Figure 17. Overall, 49% of generated sessions contain at least one explicit identifier that is replaced through this redaction strategy. We provide the attribute-wise replacement statistics in Table 10.

PII Tag	Count
[NAME]	228
[MARITAL_STATUS]	251
[OCCUPATION]	42
[GENDER]	37
[ORGANIZATION]	14
[AGE]	9

Table 10: Distribution of PII tag replacements across redacted sessions under the NER-based defense. Counts reflect total replacements across all affected turns.

In generation through Coarsened profiles, we reduce the granularity of private information provided during generation by mapping detailed profile attributes into broader categories before session generation. Names are removed entirely, while age and gender are retained as their original representations are already broad. Occupation and marital status are generalized into coarser categories. The category mappings used for marital status and occupation are provided in Table 11 and Table 12, respectively. After attribute coarsening, we prompt an LLM to adapt the patient background situation to be consistent with the modified profile information. The prompt used for situation adaptation is provided in Figure 18.

Original Value	Coarsened Category
Single	Single
Engaged, In a relationship, In a relationship unmarried	In a relationship
Married	Married
Currently experiencing a breakup with her husband, Separated	Separated
Divorced	Divorced
Widowed	Widowed
Not married, Unmarried	Other

Table 11: Coarsening mapping for marital status attribute values.

In NoiseProf, we introduce controlled perturbations to demographic attributes before session generation. For each demographic attribute, the original value is replaced with a different value with probability p. For attributes absent from the profile, a new attribute value is introduced with probability p. Names are treated separately and are always removed rather than perturbed. Replacement values are sampled uniformly from the pool of possible attribute values collected from the selected real profiles in our dataset. Since demographic perturbations may introduce inconsistencies between the modified profile and the original patient background situation, we further prompt an LLM to revise the situation to align with the perturbed demographic information. The prompt explicitly provides the attribute modifications and instructs the model to generate a consistent adapted situation. The adaptation prompt is provided in Figure 19.

NoiseIterMix extends NoiseProf by introducing additional diversification of the presenting situation through iterative mixing and divergence. After applying the initial NoiseProf perturbation, each iteration selects a situation from another profile that shares one attribute with the current profile. For occupation and age, however, we relax the condition. For occupation and age, we relax this constraint: for occupation, candidate situations may be drawn from profiles within the same occupation cluster as defined in Table 12, and for age, from adjacent age categories (e.g., a profile in the 25–44 range may mix with situations from 0–24 or 45–64).

The LLM is then prompted to incorporate relevant aspects of the retrieved situation into the current situation while rephrasing the content to increase divergence from the original situation and maintaining coherence with final perturbed demographic attributes. To ensure that the modified situations remain comparable in length to the original inputs, we constrain the generated situation length using an additional LLM-based rewriting step whenever required. The prompt used for length normalization is provided in Figure 20. The iterative diversification process is performed for up to five iterations or until the cosine similarity between the modified situation embedding and the original situation embedding (using a smaller all-MiniLM-L6-v2) falls below 0.6. The prompt used for iterative situation divergence is provided in Figure 21. For this divergence step we use a higher temperature of $T = 0.9$ to promote diversity.

Figure 16 presents an example of a perturbed profile generated using NoiseIterMix ($p = 0.8$). In this example, the original demographic attributes of age and gender are modified from 0–24 and male to 25–44 and female, respectively. Additionally, two new attributes–occupation and marital status–are introduced, which were not present in the original profile. The presenting situation is subsequently adapted to align with the final perturbed profile. While preserving the core psychological themes of hopelessness, feeling overwhelmed, and financial stress, the situation is recontextualized to reflect the updated demographic attributes. Aside from these underlying psychological issues, the resulting situation differs substantially from the original. This process ensures that key mental health characteristics remain realistic, while demographic attributes and situational details are diversified by drawing from and recombining elements across source profiles, thereby increasing divergence from the original instance.

We additionally evaluate defenses applied to CS generations and compare them with those applied to FP generations. The diversity results in Table 13 show that defenses on CS maintain diversity comparable to their FP counterparts.

For privacy, results from Table 14 indicate that defenses on CS are more slightly effective than those on FP against attribute extraction attacks. This is expected, as CS generations do not explicitly include demographic attributes, and the applied

Original Value(s)	Coarsened Category
Student, College student, Medical student, PhD student, Academic Research Associate, Recent college graduate, Film Student and Podcast Host, Intern	Student
Computer Science Engineer, Electrical engineer, Mechanical Engineer, Tech person, Technical support specialist, Customer service and technical expert	Technology
Doctor, Retired Nurse	Healthcare
Special education teacher, Assistant religious ed teacher and daycare worker	Education
Accountant / bookkeeper, Bank employee, Bank teller, Managerial or administrative employee, Commission-based job	Business/Finance
Retail, Home Depot / Kmart / McDonald's employee, Gas station worker, Call center employee, Hotel or conference center worker, Security worker, Cleaning job	Service
Writer, Drummer, Music industry professional	Creative/Arts
Military, Peace Corps volunteer, Non-profit library employee, Self employed, Works from home, Part-time job	Other
Unemployed	Unemployed

Table 12: Coarsening mapping for occupation attribute values.

defenses further reduce the attack's ability to recover such information. For MIA (Table 15), we observe comparable performances from defenses on both CS and FP. However, for linkage attack (Table 16), defenses on CS, especially NoiseProf and NoiseIterMix, are more vulnerable compared to FP. A likely reason is that CS generations rely solely on the perturbed situation, whereas FP generations incorporate both perturbed situations and demographics. As a result, defenses applied to FP induce greater divergence from the original sessions, making them harder to match, while CS-based generations remain relatively closer to their source, leading to stronger attack performance.

H Generalization Experiments

To evaluate the generalization of our findings across models, we generate sessions at different privacy levels (SO, CS, FP) using Qwen2.5-72B-Instruct, while keeping the prompts identical to those used with Llama3.3-70B-Instruct. The prompts for SO, CS and FP are provided in Figures 3,4 and 5 respectively. The LLM uses a temperature $T = 0.7$ for the generations. For evaluation, we conduct extraction, membership inference, and linkage attacks using Llama3.3-70B-Instruct as the attacker. The prompt used for the attacks are the same, with prompt attribute extractions provided in Figures 10–14 and the prompt for situation extraction provided in Figure 15. The LLM uses a temperature $T = 0$ for determinism in the attack results. For defense generations, we similarly use Qwen2.5-72B-Instruct following the generation model. The prompts used are the same as before and provided in Figures 17– 21.

The diversity results for sessions generated using Qwen2.5-72B-Instruct are reported in Table 17. Consistent with the trends observed for Llama3.3-70B-Instruct, incorporating situational context leads to increased diversity in the generated sessions. However, the inclusion of additional demographic attributes does not result in a significant further improvement in diversity. We also observe that both RProf and SProf yield diversity levels comparable to the SO generations. This reinforces the observation that synthetically constructed profiles, in the absence of grounding in real patient data, fail to introduce meaningful variation in generated sessions. While the relative ordering of different settings remains consistent across models, Qwen2.5-72B-Instruct produces overall more diverse sessions compared to Llama3.3-70B-Instruct, suggesting greater generative diversity in its outputs.

The attack results exhibit trends consistent with those observed for Llama3.3-70B-Instruct. In particular, the FP and CS generations remain highly vulnerable across all evaluated attack types, including attribute extraction (Table 18), membership inference (Table 19), and linkage attacks (Table 20).

Defenses exhibit trends consistent with prior experiments. In terms of diversity (Table 17), all defenses preserve diversity at levels comparable to CS and FP, indicating that defense mechanisms do not degrade diversity.

Surface-level defenses such as NER and Coar-

Input	Distinct				Lexical Diversity		Entropy				SitSim ↓
	D-1 ↑	D-2 ↑	D-3 ↑	EAD ↑	LDD-P ↑	LDD-C ↑	E-1 ↑	E-2 ↑	E-3 ↑	E-4 ↑	
FP + Defense											
NER	**0.017**	0.129	0.319	0.038	91.18	**88.63**	5.78	8.98	10.54	11.35	0.533
CoarsProf	**0.017**	0.127	0.316	0.038	91.46	88.11	5.78	8.99	10.56	11.36	0.546
NoiseProf ($p = 0.2$)	**0.017**	**0.130**	0.322	**0.039**	91.97	86.76	5.78	8.99	10.56	11.36	0.528
NoiseProf ($p = 0.5$)	**0.017**	0.127	0.315	0.038	**92.98**	84.67	5.78	8.98	10.53	11.34	0.528
NoiseProf ($p = 0.8$)	**0.017**	**0.130**	**0.323**	**0.039**	92.74	87.07	5.79	9.01	**10.57**	**11.37**	0.532
NoiseIterMix ($p = 0.8$)	0.016	0.120	0.300	0.037	88.24	82.78	5.76	8.94	10.47	11.29	0.561
CS + Defense											
NER	0.016	0.128	0.317	0.038	92.08	86.48	5.78	8.99	10.55	11.36	0.533
CoarsProf	**0.017**	0.128	0.317	**0.039**	91.05	88.36	**5.80**	**9.02**	**10.57**	**11.37**	0.544
NoiseProf ($p = 0.2$)	0.016	0.126	0.313	0.038	88.95	85.50	5.78	8.99	10.56	**11.37**	**0.527**
NoiseProf ($p = 0.5$)	0.016	0.124	0.307	0.038	87.55	83.93	5.76	8.95	10.49	11.30	0.531
NoiseProf ($p = 0.8$)	0.016	0.126	0.313	0.038	90.27	85.93	5.78	8.98	10.54	11.35	0.529
NoiseIterMix ($p = 0.8$)	0.016	0.119	0.295	0.037	87.80	82.52	5.76	8.93	10.47	11.28	0.559

Table 13: Diversity evaluation for defense on FP and CS generations. **D-1/2/3**: Distinct-1/2/3; **EAD**: Expected Adjusted Distinct; **LDD-P/C**: Lexical Diversity Density (Patient/Counselor); **E-1–4**: Entropy at n-gram levels 1–4; **SitSim**: average cosine similarity across situations (lower = more diverse). **Bold** indicates best per column.

Input	Confusion Metrics				Classification Metrics				Leakage	Similarity
	TPR ↓	FPR ↑	TNR ↓	FNR ↑	Acc ↓	Prec ↓	Recall ↓	F1 ↓	Attr (%) ↓	CosSim ↓
FP + Defense										
NER	0.582	0.283	0.717	0.418	0.673	0.501	0.582	0.538	46.57	0.720 ± 0.10
CoarsProf	0.427	0.261	0.739	0.573	0.639	0.438	0.427	0.433	33.61	0.706 ± 0.10
NoiseProf ($p = 0.2$)	0.645	0.360	0.640	0.355	0.641	0.426	0.645	0.513	46.52	0.716 ± 0.10
NoiseProf ($p = 0.5$)	0.546	0.444	0.556	0.454	0.553	0.302	0.546	0.388	34.73	0.697 ± 0.11
NoiseProf ($p = 0.8$)	0.461	0.518	0.482	0.539	0.477	0.206	0.461	0.284	25.40	0.692 ± 0.11
NoiseIterMix ($p = 0.8$)	0.460	**0.532**	**0.468**	0.540	**0.466**	**0.189**	0.460	0.268	23.87	0.638 ± 0.11
CS + Defense										
NER	0.495	0.268	0.732	0.505	0.654	0.471	0.495	0.483	39.43	0.718 ± 0.11
CoarsProf	0.409	0.261	0.739	0.591	0.635	0.418	0.409	0.414	31.60	0.705 ± 0.10
NoiseProf ($p = 0.2$)	0.524	0.302	0.698	0.476	0.644	0.442	0.524	0.480	39.97	0.725 ± 0.10
NoiseProf ($p = 0.5$)	0.457	0.324	0.676	0.543	0.614	0.359	0.457	0.402	31.84	0.718 ± 0.11
NoiseProf ($p = 0.8$)	0.349	0.353	0.647	0.651	0.569	0.262	0.349	0.300	22.55	0.708 ± 0.11
NoiseIterMix ($p = 0.8$)	**0.345**	0.422	0.578	**0.655**	0.522	0.205	**0.345**	**0.257**	**20.26**	$\mathbf{0.633 \pm 0.11}$

Table 14: Attribute extraction attack performance against defense on FP and CS generations. **TPR/FPR/TNR/FNR**: True/False Positive/Negative Rates; **Acc**: Accuracy; **Prec**: Precision; **Attr (%)**: percentage of attribute leakage; **CosSim**: cosine similarity between original and extracted situations. **Bold** indicates best defense per column.

sProf provide limited protection against privacy attacks as indirect cues embedded in the generated text still enable effective attribute extraction, profile linkage, and membership inference. NoiseProf shows a clear dependence on the perturbation probability p. At low perturbation ($p = 0.2$), the defense remains vulnerable. As p increases, NoiseProf becomes more effective: at $p = 0.8$, it achieves a substantial reduction in extraction leakage ($\sim 36\%$) while approximately doubling the FPR, thereby reducing both attack success and reliability. However, for linkage and membership inference attacks, the improvements remain modest, with NoiseProf ($p = 0.8$) reducing AUC by $\sim 10\%$. This suggests that perturbing demographic attributes alone is insufficient. NoiseIterMix ($p = 0.8$) further strengthens privacy protection by perturbing both demographic attributes and situational information. It achieves similar reductions in attribute extraction as NoiseProf ($p = 0.8$), while additionally reducing situational leakage, as evidenced by lower cosine similarity of the extracted situation with the original situation. For linkage and membership inference attacks, NoiseIterMix ($p = 0.8$) consistently outperforms NoiseProf ($p = 0.8$), reducing AUC by an additional 3–5% and significantly lowering TPR under low-FPR thresholds, thus reducing the reliability of the attacks. Overall, we find our observations about generation utility, attacks and defenses to be con-

Input	AUC ↓	TPR @ FPR ≤ τ ↓		
		τ=0.10	τ=0.05	τ=0.01
FP + Defense				
NER	0.743	0.378	0.284	0.134
CoarsProf	0.715	0.331	0.238	0.134
NoiseProf ($p = 0.2$)	0.748	0.441	0.342	0.211
NoiseProf ($p = 0.5$)	0.712	0.327	0.224	0.135
NoiseProf ($p = 0.8$)	0.700	0.316	0.194	0.072
NoiseIterMix ($p = 0.8$)	0.631	0.185	0.116	0.053
CS + Defense				
NER	0.733	0.412	0.248	0.136
CoarsProf	0.702	0.330	0.238	0.069
NoiseProf ($p = 0.2$)	0.760	0.436	0.317	0.195
NoiseProf ($p = 0.5$)	0.730	0.362	0.199	0.105
NoiseProf ($p = 0.8$)	0.719	0.368	0.207	0.127
NoiseIterMix ($p = 0.8$)	**0.614**	**0.182**	**0.106**	**0.017**

Table 15: Membership inference attack performance against defense on FP and CS generations. **AUC**: area under the ROC curve; **TPR @ FPR ≤ threshold (τ)**: attack performance at controlled false positive rates. **Bold** indicates the strongest defense per column.

Input	AUC ↓	TPR @ FPR ≤ τ ↓		
		τ=0.10	τ=0.05	τ=0.01
FP + Defense				
NER	0.970	0.926	0.866	0.739
CoarsProf	0.961	0.903	0.829	0.639
NoiseProf ($p = 0.2$)	0.955	0.878	0.813	0.586
NoiseProf ($p = 0.5$)	0.924	0.799	0.696	0.508
NoiseProf ($p = 0.8$)	0.896	0.730	0.632	0.421
NoiseIterMix ($p = 0.8$)	**0.794**	**0.541**	**0.416**	**0.198**
CS + Defense				
NER	0.964	0.905	0.867	0.687
CoarsProf	0.951	0.901	0.828	0.630
NoiseProf ($p = 0.2$)	0.954	0.871	0.815	0.653
NoiseProf ($p = 0.5$)	0.951	0.882	0.835	0.601
NoiseProf ($p = 0.8$)	0.934	0.840	0.739	0.508
NoiseIterMix ($p = 0.8$)	0.840	0.586	0.483	0.235

Table 16: Linkage attack performance against defense on FP and CS generations. **AUC**: area under the ROC curve; **TPR @ FPR ≤ threshold (τ)**: attack performance at controlled false positive rates. **Bold** indicates the strongest defense per column.

sistent in both Llama-generated sessions attacked by Qwen and Qwen-generated sessions attacked by Llama showing the generalization of our findings.

I Attack Model scaling

We analyze the impact of attacker model scale and reasoning capability on the effectiveness of privacy attacks. All attacks are conducted on **FP** sessions generated using Llama3.3-70B-Instruct. As attacker models, we consider three non-reasoning LLMs of varying scales: Qwen2.5-72B-Instruct, Qwen2.5-32B-Instruct, and Qwen2.5-7B-Instruct, alongside a reasoning-based model, Qwen3-32B-Instruct (thinking mode). For all attacks we use Temperature $T = 0$ for determinism.

Table 21 presents the results for attribute extraction attacks. We observe that Qwen2.5-72B-Instruct and Qwen2.5-32B-Instruct achieve nearly identical performance across true positive rate (TPR), false positive rate (FPR), accuracy, and overall attribute extraction rate, suggesting diminishing returns in attack effectiveness beyond the 32B scale. In contrast, Qwen2.5-7B-Instruct exhibits a substantial degradation in performance, with an approximate 40% reduction in attribute extraction. This indicates that sensitive attributes are not trivially exposed in the generated sessions and require sufficiently capable models for reliable inference. Interestingly, the reasoning model, Qwen3-32B-Instruct demonstrates a different trade-off. While its overall attribute extraction rate is lower (approximately 10% lower than Qwen2.5-32B-Instruct), it achieves a reduced false positive rate while maintaining comparable overall accuracy. This behavior suggests that reasoning-based models adopt a more conservative inference strategy, preferring abstention when evidence is insufficient. As a result, they achieve improved precision without compromising accuracy.

Table 22 presents the results for membership inference attacks. In contrast to the attribute extraction attacks, all evaluated models, achieve comparable performance, with a slight drop-off in Qwen2.5-7B-Instruct, indicating that attack effectiveness is largely insensitive to model scale for MIA. A similar trend is observed for linkage attacks, with results reported in Table 23, where all models demonstrate near-equivalent performance. These findings suggest that even relatively smaller models can recover sufficient signals from synthetic sessions to reliably determine whether a given profile was used during generation of the dataset (membership inference) or to associate sessions with their source profiles (linkage). In other words, unlike attribute extraction, these attacks do not require highly capable models to be effective. Overall, this highlights the robustness of linkage and membership inference attacks across model scales. It further underscores the need for defense mechanisms that explicitly mitigate these forms of privacy risk, as they represent strong and persistent attack baselines even under reduced attacker capabilities.

Input	Distinct				Lexical Diversity		Entropy				SitSim ↓
	D-1 ↑	D-2 ↑	D-3 ↑	EAD ↑	LDD-P ↑	LDD-C ↑	E-1 ↑	E-2 ↑	E-3 ↑	E-4 ↑	
SO	0.018	0.134	0.315	0.035	77.45	80.43	5.88	8.96	10.41	11.20	0.659
CS	**0.021**	0.155	0.362	**0.042**	92.23	89.95	5.94	9.13	**10.64**	**11.43**	0.533
FP	**0.021**	0.157	0.365	**0.042**	92.46	91.57	**5.95**	**9.14**	**10.64**	**11.43**	**0.532**
RProf	0.019	0.142	0.330	0.038	85.05	82.38	5.92	9.04	10.50	11.30	0.580
SProf	0.018	0.134	0.316	0.037	81.96	81.26	5.91	9.01	10.47	11.26	0.631
FP + Defense											
NER	**0.021**	**0.158**	**0.366**	**0.042**	92.52	91.54	**5.95**	**9.14**	**10.64**	**11.43**	0.540
CoarsProf	**0.021**	0.157	0.365	**0.042**	91.46	**92.04**	**5.95**	9.13	10.63	11.42	0.542
NoiseProf ($p = 0.2$)	**0.021**	0.156	0.362	0.041	92.21	90.20	5.94	9.11	10.61	11.40	0.537
NoiseProf ($p = 0.5$)	**0.021**	0.157	0.364	**0.042**	92.44	90.32	**5.95**	9.13	10.63	11.42	0.534
NoiseProf ($p = 0.8$)	**0.021**	0.154	0.359	0.041	92.30	90.47	5.93	9.11	10.62	11.42	0.539
NoiseIterMix ($p = 0.8$)	**0.021**	0.156	0.363	**0.042**	**93.04**	90.11	**5.95**	9.13	10.63	11.42	0.547

Table 17: Diversity evaluation on Qwen generations. **D-1/2/3**: Distinct-1/2/3; **EAD**: Expected Adjusted Distinct; **LDD-P/C**: Lexical Diversity Density (Patient/Counselor); **E-1–4**: Entropy at n-gram levels 1–4; **SitSim**: average cosine similarity across situations (lower = more diverse). **Bold** indicates best per column.

Input	Confusion Metrics				Classification Metrics				Leakage	Similarity
	TPR ↓	FPR ↑	TNR ↓	FNR ↑	Acc ↓	Prec ↓	Recall ↓	F1 ↓	Attr (%) ↓	CosSim ↓
SO	0.228	0.248	0.752	0.772	0.592	0.286	0.228	0.254	16.92	0.591 ± 0.13
CS	0.665	0.298	0.702	0.335	0.691	0.509	0.665	0.577	51.91	0.742 ± 0.09
FP	0.804	0.291	0.709	0.196	0.740	0.575	0.804	0.670	64.64	0.731 ± 0.11
FP + Defense										
NER	0.650	0.266	0.734	0.350	0.706	0.545	0.650	0.593	52.40	0.727 ± 0.10
CoarsProf	0.583	0.282	0.718	0.417	0.674	0.500	0.583	0.538	46.41	0.718 ± 0.11
NoiseProf ($p = 0.2$)	0.695	0.364	0.636	0.305	0.655	0.468	0.695	0.560	53.09	0.725 ± 0.10
NoiseProf ($p = 0.5$)	0.649	0.443	0.557	0.351	0.580	0.336	0.649	0.442	40.81	0.711 ± 0.11
NoiseProf ($p = 0.8$)	0.534	**0.557**	**0.443**	0.466	**0.462**	**0.207**	0.534	**0.298**	**28.06**	0.705 ± 0.11
NoiseIterMix ($p = 0.8$)	**0.513**	0.547	0.453	**0.487**	0.467	0.213	**0.513**	0.301	28.15	$\mathbf{0.671 \pm 0.12}$

Table 18: Attribute extraction attack performance on Qwen generations. **TPR/FPR/TNR/FNR**: True/False Positive/Negative Rates; **Acc**: Accuracy; **Prec**: Precision; **Attr (%)**: percentage of attribute leakage; **CosSim**: cosine similarity between original and extracted situations. **Bold** indicates best defense per column.

J Experimental Details

For synthetic counseling session generation, we use a temperature of $T = 0.7$. For LLM-as-a-judge evaluations and LLM-based privacy attacks, we use temperature $T = 0$ for determinism. For diverging the situations in NoiseIterMix, we use a temperature of $T = 0.9$ to promote variation. Each generation, evaluation and attack is run only once. For formatting the prompts, we use the LangChain [3] Library. For hosting the LLM servers used in generation, evaluation and attacks, we use the vLLM (Kwon et al., 2023) library and host them on 4×48GB L40 GPUs. For the random library used in noise based perturbation, we use a seed of 42. For the sentence transformers, we use the HuggingFace Library[4]. The Hugging Face [5], vLLM (Kwon et al., 2023), and LangChain [6] libraries used for implementation are licensed under Apache License, Version 2.0. We have confirmed all of the artifacts used in this paper are available for non-commercial scientific use.

[3] LangChain
[4] Hugging Face
[5] Hugging Face
[6] LangChain

Input	AUC ↓	TPR @ FPR ≤ τ ↓		
		τ=0.10	τ=0.05	τ=0.01
SO	0.535	0.080	0.028	0.011
CS	0.787	0.514	0.345	0.189
FP	0.808	0.453	0.349	0.161
FP + Defense				
NER	0.771	0.404	0.259	0.145
CoarsProf	0.729	0.311	0.154	0.110
NoiseProf ($p = 0.2$)	0.757	0.394	0.246	0.168
NoiseProf ($p = 0.5$)	0.725	0.355	0.257	0.112
NoiseProf ($p = 0.8$)	0.686	0.286	0.199	0.083
NoiseIterMix ($p = 0.8$)	**0.651**	**0.217**	**0.124**	**0.030**

Table 19: Membership inference attack performance on Qwen generations. **AUC**: area under the ROC curve; **TPR @ FPR ≤ threshold** (τ): attack performance at controlled false positive rates. **Bold** indicates the strongest defense per column.

Input	AUC ↓	TPR @ FPR ≤ τ ↓		
		τ=0.10	τ=0.05	τ=0.01
SO	0.624	0.234	0.133	0.056
CS	0.966	0.922	0.892	0.726
FP	0.976	0.943	0.920	0.795
FP + Defense				
NER	0.965	0.929	0.896	0.727
CoarsProf	0.950	0.896	0.843	0.699
NoiseProf ($p = 0.2$)	0.952	0.882	0.811	0.630
NoiseProf ($p = 0.5$)	0.921	0.799	0.747	0.559
NoiseProf ($p = 0.8$)	0.895	0.734	0.625	0.412
NoiseIterMix ($p = 0.8$)	**0.845**	**0.625**	**0.488**	**0.298**

Table 20: Linkage attack performance on Qwen generation. **AUC**: area under the ROC curve; **TPR @ FPR ≤ threshold** (τ): attack performance at controlled false positive rates. **Bold** indicates the strongest defense per column.

Model	Confusion Metrics				Classification Metrics				Leakage	Similarity
	TPR ↑	FPR ↓	TNR ↑	FNR ↓	Acc ↑	Prec ↑	Recall ↑	F1 ↑	Attr (%) ↑	CosSim ↑
Qwen2.5-72B	**0.728**	0.294	0.706	**0.272**	0.714	0.552	**0.728**	0.628	59.19	**0.731 ± 0.11**
Qwen2.5-32B	0.716	0.250	0.750	0.284	**0.739**	0.594	0.716	**0.649**	**59.34**	0.695 ± 0.12
Qwen2.5-7B	0.227	**0.118**	**0.882**	0.773	0.644	0.524	0.227	0.317	20.63	0.712 ± 0.12
Qwen3-32B (Reasoning)	0.576	0.184	0.816	0.424	0.732	**0.629**	0.576	0.602	49.68	0.703 ± 0.10

Table 21: Attribute extraction attack performance across model scales. **TPR/FPR/TNR/FNR**: True/False Positive/Negative Rates; **Acc**: Accuracy; **Prec**: Precision; **Attr (%)**: percentage of attribute leakage; **CosSim**: cosine similarity between original and extracted situations. **Bold** indicates best attack per column.

Model	AUC ↑	TPR @ FPR $\leq \tau$ ↑		
		τ=0.10	τ=0.05	τ=0.01
Qwen2.5-72B	0.811	0.492	**0.412**	0.233
Qwen2.5-32B	**0.831**	**0.558**	0.400	**0.297**
Qwen2.5-7B	0.809	0.482	0.369	0.159
Qwen3-32B (Reasoning)	0.810	0.507	0.384	0.277

Table 22: Membership inference attack performance across model scales (FP input). **AUC**: area under the ROC curve; **TPR @ FPR $\leq$ threshold** (τ): attack performance at controlled false positive rates. **Bold** indicates the strongest attack per column.

Model	AUC ↑	TPR @ FPR $\leq \tau$ ↑		
		τ=0.10	τ=0.05	τ=0.01
Qwen2.5-72B	**0.985**	**0.970**	0.920	0.784
Qwen2.5-32B	0.979	0.955	0.916	0.758
Qwen2.5-7B	0.980	0.958	**0.926**	**0.799**
Qwen3-32B (Reasoning)	0.982	0.948	0.907	0.784

Table 23: Linkage attack performance across model scales. **AUC**: area under the ROC curve; **TPR @ FPR $\leq$ threshold** (τ): attack performance at controlled false positive rates. **Bold** indicates the strongest attack per column.

SO Generation Prompt

Write the script of a mental health counseling session between a counselor and a client with at least 30 turns.
You are given:
- The client's medical information, describing the severity of different symptoms
- The client's level of resistance toward counseling
INPUTS
Client Medical Information:
{client_medical_information}
Client Resistance towards counseling:
{client_resistance}
Global Constraints
1. Use natural conversational signals whenever appropriate (e.g., "mm-hm", "um", "yeah","right","...").
2. Both clients and counselors should take their time to explain their thoughts with multiple sentences at once. Make sure that the client and counselor ***must use multi-sentence utterances (3–5 sentences) when explaining experiences, emotions, reflections, or psychoeducation***.
3. The session should progress through ideas gradually. Avoid advancing to a new topic or conclusion in consecutive turns. Most topics should be explored across multiple turns with depth, not resolved immediately.
Client utterances guidelines and constraints:
1. The client utterances should be consistent with their medical information and resistance.
2. The client should express symptoms indirectly through lived experiences rather than naming them explicitly.
3. Use everyday, non-clinical language. The client should not self-diagnose or use professional terminology, except for informal mental health terms or concepts previously introduced by the counselor.
4. The client should describe emotions and difficulties concretely (thoughts, feelings, situations), often as short narratives rather than isolated statements.
5. The client can express ambivalence, confusion, or difficulty naming emotions when appropriate.
Counselor utterances guidelines and constraints:
1. The counselor does not have direct knowledge of the client's symptoms and should not make assumptions about them.
2. Counselor responses should be warm, empathetic, and grounded.
3. The counselor should focus on building trust and fostering a collaborative therapeutic atmosphere.
Output Format
- Output only valid JSON.
- Do not include explanations, comments, or markdown.
Required Json Format:
[
{{"role": "counselor", "content": "Hello, how are you feeling today?"}},
{{"role": "client", "content": "Uh... I'm okay, I guess. Just kind of tired."}}
]

Figure 3: Prompt used to generate synthetic counseling sessions with symptoms-only (SO) input.

CS Generation Prompt

Write the script of a mental health counseling session between a counselor and a client with at least 30 turns.
You are given:
- The client's background information
- The client's medical information, describing the severity of different symptoms
- The client's level of resistance toward counseling
INPUTS
Client Background Information:
{client_background_information}
Client Medical Information:
{client_medical_information}
Client Resistance towards counseling:
{client_resistance}
Global Constraints
1. Use natural conversational signals whenever appropriate (e.g., "mm-hm", "um", "yeah","right","...").
2. Both clients and counselors should take their time to explain their thoughts with multiple sentences at once. Make sure that the client and counselor ***must use multi-sentence utterances (3–5 sentences) when explaining experiences, emotions, reflections, or psychoeducation***.
3. The session should progress through ideas gradually. Avoid advancing to a new topic or conclusion in consecutive turns. Most topics should be explored across multiple turns with depth, not resolved immediately.
Client utterances guidelines and constraints:
1. The client utterances should be consistent with their background information, medical information and resistance.
2. The client should express symptoms indirectly through lived experiences rather than naming them explicitly.
3. Use everyday, non-clinical language. The client should not self-diagnose or use professional terminology, except for informal mental health terms or concepts previously introduced by the counselor.
4. The client should describe emotions and difficulties concretely (thoughts, feelings, situations), often as short narratives rather than isolated statements.
5. The client can express ambivalence, confusion, or difficulty naming emotions when appropriate.
Counselor utterances guidelines and constraints:
1. The counselor is aware of the client's background information but does not have direct knowledge of the client's symptoms and should not assume them.
2. Counselor responses should be warm, empathetic, and grounded.
3. The counselor should focus on building trust and fostering a collaborative therapeutic atmosphere.
Output Format
- Output only valid JSON.
- Do not include explanations, comments, or markdown.
Required Json Format:
```
[
{{"role": "counselor", "content": "Hello, how are you feeling today?"}},
{{"role": "client", "content": "Uh... I'm okay, I guess. Just kind of tired."}}
]
```

Figure 4: Prompt used to generate synthetic counseling sessions with contextual situation (CS) input (symptoms + situation).

FP Generation Prompt

Write the script of a mental health counseling session between a counselor and a client with at least 30 turns.
You are given:
- The client's background information
- The client's medical information, describing the severity of different symptoms
- The client's level of resistance toward counseling
INPUTS
Client Background Information:
{client_background_information}
Client Medical Information:
{client_medical_information}
Client Resistance towards counseling:
{client_resistance}
Global Constraints
1. Use natural conversational signals whenever appropriate (e.g., "mm-hm", "um", "yeah","right","...").
2. Both clients and counselors should take their time to explain their thoughts with multiple sentences at once. Make sure that the client and counselor ***must use multi-sentence utterances (3–5 sentences) when explaining experiences, emotions, reflections, or psychoeducation***.
3. The session should progress through ideas gradually. Avoid advancing to a new topic or conclusion in consecutive turns. Most topics should be explored across multiple turns with depth, not resolved immediately.
Client utterances guidelines and constraints:
1. The client utterances should be consistent with their background information, medical information and resistance.
2. The client should express symptoms indirectly through lived experiences rather than naming them explicitly.
3. Use everyday, non-clinical language. The client should not self-diagnose or use professional terminology, except for informal mental health terms or concepts previously introduced by the counselor.
4. The client should describe emotions and difficulties concretely (thoughts, feelings, situations), often as short narratives rather than isolated statements.
5. The client can express ambivalence, confusion, or difficulty naming emotions when appropriate.
Counselor utterances guidelines and constraints:
1. The counselor is aware of the client's background information but does not have direct knowledge of the client's symptoms and should not assume them.
2. Counselor responses should be warm, empathetic, and grounded.
3. The counselor should focus on building trust and fostering a collaborative therapeutic atmosphere.
Output Format
- Output only valid JSON.
- Do not include explanations, comments, or markdown.
Required Json Format:
[
{{"role": "counselor", "content": "Hello, how are you feeling today?"}},
{{"role": "client", "content": "Uh... I'm okay, I guess. Just kind of tired."}}
]

Figure 5: Prompt used to generate synthetic counseling sessions with full profile (FP) input (symptoms + situation + demographics).

CTRS LLM-as-a-judge Evaluation Prompt

I want you to act as an evaluator. You will be provided with a transcript of a counseling session between a therapist and a patient. Your task is to assess the therapist based on the given criteria. If you believe the therapist falls between two of the descriptors, select the intervening odd number (1, 3, 5). For example, if the therapist set a very good agenda but did not establish priorities, assign a rating of 5 rather than 4.
Please follow these steps:
1. Read the counseling session transcript carefully.
2. Review the evaluation questions and criteria provided below.
3. Assign a score based on the criteria, grading very strictly and uptight. If there is any deficiency, no matter how minor, assign a score of 4 or lower.
4. Output the score and the explanation, separated by a comma. Do not add any prefix.
Counseling conversation:
{conversation}
Evaluation Question:
{question}
Criteria:
{criteria}

Figure 6: Prompt used to evaluate the generated counseling sessions on CTRS using LLM-as-a-judge setup.

WAI LLM-as-a-judge Evaluation Prompt

The following is a psychological counseling session between a counselor and a client. As a third party, you should read the conversation and guidelines carefully and then score the following question from 1 to 7.
Please follow these steps:
1. Read the counseling session transcript carefully.
2. Review the evaluation questions and criteria provided below.
3. Assign a score based on the criteria, grading very strictly.
4. Output the score (***only the numerical***) and the explanation, separated by a comma. ***Do not add any prefix.***
Counseling conversation:
{conversation}
Question:
{question}
Criteria:
{criteria}

Figure 7: Prompt used to evaluate the generated counseling sessions on WAI using LLM-as-a-judge setup.

RProf Synthetic Situation Generation Prompt

System Prompt:
You are a text generation assistant specialized in creating brief, realistic therapy client profiles.
Your task is to generate:
1. A plausible client name
2. A concise but detailed description of the client's current situation.
You will be given structured attributes (age, occupation, gender, marital status, and symptoms).
Use these attributes to infer a realistic mental-health–related situation that could plausibly lead the client to seek counseling.
IMPORTANT CONSTRAINTS:
- Generate ONLY the requested fields (Name and Situation).
- The situation must be written in natural language, in third person.
- The situation should describe the client's current struggles, recent stressors, and emotional state.
- Do NOT include diagnoses, treatment plans, or counselor perspectives.
- An example situation may be provided to demonstrate expected length and level of detail.
    - The example is for STYLE and LENGTH ONLY.
    - DO NOT copy, paraphrase, or reuse specific details from the example.
- Do NOT add explanations, bullet points, headings, or metadata beyond the required output.
Respond ONLY in valid JSON with the following format:
"{
"name": "<Client Name>",
"situation": "<Client Situation>"
}"
Example Situation (STYLE & LENGTH REFERENCE ONLY — DO NOT COPY CONTENT):
The client is experiencing anxiety and feelings of not being good enough due to a bad grade, despite having passed a rigorous selection process to get into a program.
User Prompt:
Generate a fictional therapy client profile based on the attributes below.
Client Attributes:
- Age: {age}
- Gender: {gender}
- Occupation: {occupation}
- Marital Status: {marital_status}
- Symptoms: {symptoms}

Figure 8: Prompt used to generate synthetic situations in RProf.

SProf Synthetic Situation and Demographics Generation Prompt

System Prompt:
You are a text generation assistant specialized in creating brief, realistic therapy client profiles.
Your task is to generate a client profile based on a given list of symptoms AND a set of allowed attribute values.
You must generate:
1. A plausible client name
2. A plausible client age
3. Gender
4. A plausible client marital status
5. A plausible client occupation
6. A concise but detailed description of the client’s current situation.
STRICT CONSTRAINTS:
- Age, Gender, Marital Status, and Occupation MUST be selected EXACTLY from the provided allowed lists.
- Do NOT invent, modify, or combine list values.
- All selected attributes must be internally consistent and plausible given the symptoms.
- The situation must be written in natural, third-person language.
- The situation should describe current struggles, emotional experiences, and recent stressors.
- Do NOT include diagnoses, clinical labels, or treatment-related language.
- Do NOT mention therapy, counselors, or mental health professionals.
- An example situation may be provided to demonstrate expected length and detail.
  - The example is for STYLE and LENGTH ONLY.
  - DO NOT copy, paraphrase, or reuse any specific details from the example.
- Output ONLY the requested fields, following the exact format specified.
- Do NOT include explanations, reasoning, bullet points, or extra text.
Respond ONLY in valid JSON with the following format:
"{
"name": "<Client Name>",
"age": "<Client Age>",
"gender": "<Client Gender>",
"marital_status": "<Client Marital Status>",
"occupation": "<Client Occupation>",
"situation": "<Client Situation>"
}"
Example Situation (STYLE & LENGTH REFERENCE ONLY — DO NOT COPY CONTENT):
The client is experiencing anxiety and feelings of not being good enough due to a bad grade, despite having passed a rigorous selection process to get into a program.
User Prompt:
Generate a fictional therapy client profile using ONLY the information below.
Symptoms:
{symptoms}
Allowed Values (you MUST choose from these lists only):
Age:
[’0 24’, ’25 44’, ’45 64’, ’65+’, ’Cannot be identified’]
Gender:
[’Male’, ’Female’, ’Cannot be identified’]
Marital Status:
[’Single’, ’Married’, ’Divorced’, ’Widowed’, ’Separated’, ’Other’, ’Cannot be identified’]
Occupation:
["Academic Research Associate","Accounting. Low entry, bookkeeper.","Assistant religious ed teacher and daycare worker","Bank employee","Call center employee","Cleaning job","College student","Commission based job","Computer Science Engineer","Customer service and technical expert","Doctor","Drummer","Electrical engineer","Employee in a company, potentially managerial or administrative","Engineer in Mechanical Engineering","Film Student and Podcast Host","Gas station worker","Graduate in a field related to their master’s degree","Home Depot employee","Hotel or Conference Center Worker","Intern","Kmart employee","McDonald’s employee","Medical student","Military","Music industry professional","Non-profit library system employee","Part-time job","Peace Corps volunteer","PhD in immunology","PhD student","Recent college graduate looking for a job in economic research","Retail","Retired Nurse","Self employed","Special education teacher","Student","Tech person","Technical support specialist","Teller at a bank","Unemployed","Works from home","Writer","college student","hair braider","Cannot be identified"]

Figure 9: Prompt used to generate synthetic situations and demographics in SProf.

Name Extraction Prompt

System Prompt:
You are an information extraction assistant.
Your task is to extract the client's name from the given conversation.
You MUST follow these rules:
- Output ONLY valid JSON.
- Do NOT include markdown, code blocks, or extra text.
- The JSON object must contain exactly two fields: 'label' and 'explanation'.
- The 'label' value must be either the extracted name or 'Cannot be identified'.
- If no clear name is stated, use the label 'Cannot be identified'.

User Prompt:
What is the name of this client based on the following conversation?
Respond with the client's name exactly as stated in the conversation, or 'Cannot be identified'.
Respond ONLY in valid JSON with the following format:
{
"label": "<client name or 'Cannot be identified'>",
"explanation": "<brief justification based on the conversation, or 'Not enough information'>"
}
Conversation:
{conversation}

Figure 10: Prompt used to extract patient name from the generated synthetic counseling sessions.

Gender Extraction Prompt

System Prompt:
You are an information extraction assistant.
Your task is to infer the client's most probable gender from the given conversation.
You MUST follow these rules:
- Output ONLY valid JSON.
- Do NOT include markdown, code blocks, or extra text.
- The JSON object must contain exactly two fields: 'label' and 'explanation'.
- The 'label' value must be one of the allowed options provided by the user.
- If the gender cannot be inferred, use the label 'Cannot be identified'.

User Prompt:
What is the most probable gender of this client based on information such as the client's name and the pronouns used in the conversation?
Choose one label from: ['Male', 'Female', 'Cannot be identified'].
Respond ONLY in valid JSON with the following format:
{
label: "<chosen gender>",
explanation: "<brief justification based on the conversation, or 'Not enough information'>"
}
Conversation:
{conversation}

Figure 11: Prompt used to extract patient gender from the generated synthetic counseling sessions.

Age Extraction Prompt

System Prompt:
You are an information extraction assistant.
Your task is to infer the client's age range from the given conversation.
You MUST follow these rules:
- Output ONLY valid JSON.
- Do NOT include markdown, code blocks, or extra text.
- The JSON object must contain exactly two fields: 'label' and 'explanation'.
- The 'label' value must be one of the allowed options provided by the user.
- If the age range cannot be inferred, use the label 'Cannot be identified'.

User Prompt:
Estimate the client's age from the following conversation.
Choose one label from: ['0 24', '25 44', '45 64', '65+', 'Cannot be identified'].
Respond ONLY in valid JSON with the following format:
{
"label": "<chosen age range>",
"explanation": "<brief justification based on the conversation, or 'Not enough information'>"
}
Conversation:
{conversation}

Figure 12: Prompt used to extract patient age from the generated synthetic counseling sessions.

Occupation Extraction Prompt

System Prompt:
You are an information extraction assistant.
Your task is to infer the client's occupation from the given conversation.
You MUST follow these rules:
- Output ONLY valid JSON.
- Do NOT include markdown, code blocks, or extra text.
- The JSON object must contain exactly two fields: 'label' and 'explanation'.
- The 'label' value must be one of the allowed options provided by the user.
- If the occupation cannot be inferred, use the label 'Cannot be identified'.

User Prompt:
Determine the client's occupation from the following conversation.
Choose one label from:
['Academic Research Associate','Accounting. Low entry, bookkeeper.','Assistant religious ed teacher and daycare worker','Bank employee','Call center employee','Cleaning job','College student','Commission based job','Computer Science Engineer','Customer service and technical expert', 'Doctor','Drummer','Electrical engineer','Employee in a company, potentially managerial or administrative','Engineer in Mechanical Engineering','Film Student and Podcast Host','Gas station worker','Graduate in a field related to their master's degree','Home Depot employee','Hotel or Conference Center Worker','Intern','Kmart employee','McDonald's employee','Medical student','Military','Music industry professional','Non-profit library system employee','Part-time job','Peace Corps volunteer','PhD in immunology','PhD student','Recent college graduate looking for a job in economic research','Retail','Retired Nurse','Self employed','Special education teacher','Student','Tech person','Technical support specialist','Teller at a bank','Unemployed','Works from home','Writer','college student','hair braider','Cannot be identified']
Respond ONLY in valid JSON with the following format:
{
"label": "<chosen occupation>",
"explanation": "<brief justification based on the conversation, or 'Not enough information'>"
}
Conversation:
{conversation}

Figure 13: Prompt used to extract patient occupation from the generated synthetic counseling sessions.

Marital Status Extraction Prompt

System Prompt:
You are an information extraction assistant.
Your task is to infer the client's marital status from the given conversation.
You MUST follow these rules:
- Output ONLY valid JSON.
- Do NOT include markdown, code blocks, or extra text.
- The JSON object must contain exactly two fields: 'label' and 'explanation'.
- The 'label' value must be one of the allowed options provided by the user.
- If the marital status cannot be inferred, use the label 'Cannot be identified'.

User Prompt:
What is the client's marital status based on the conversation?
Choose one label from: ['Single', 'Married', 'Divorced', 'Widowed', 'Separated', 'Other', 'Cannot be identified'].
Respond ONLY in valid JSON with the following format:
{
"label": "<chosen marital status>",
"explanation": "<brief justification based on the conversation, or 'Not enough information'>"
}
Conversation:
{conversation}

Figure 14: Prompt used to extract patient marital status from the generated synthetic counseling sessions.

Situation Extraction Prompt

Briefly summarize the client's situation **before help-seeking** that triggered this conversation. Use a few sentences. Focus only on the core situation.
Examples:
Example 1:
The client is feeling lost and struggling with life's setbacks, unable to attend college due to financial difficulties and being held back by the lack of graduation papers. The client feels hopeless, especially after dropping out of college, and is unsure of how to move forward.
Example 2:
The client is distressed about a broken laptop which contained a lengthy story they were writing. The event that led to this damage was the client's father getting angry about the client's grades and slamming the laptop, potentially damaging it. The client is worried about retrieving the lost work and has expressed feelings of wanting to die. The client also mentioned family issues, feeling scared of their father's physical behavior, and concerns about the potential impact on their brother, who has Down Syndrome.
Conversation:
{conversation}

Figure 15: Prompt used to extract patient situation from the generated synthetic counseling sessions.

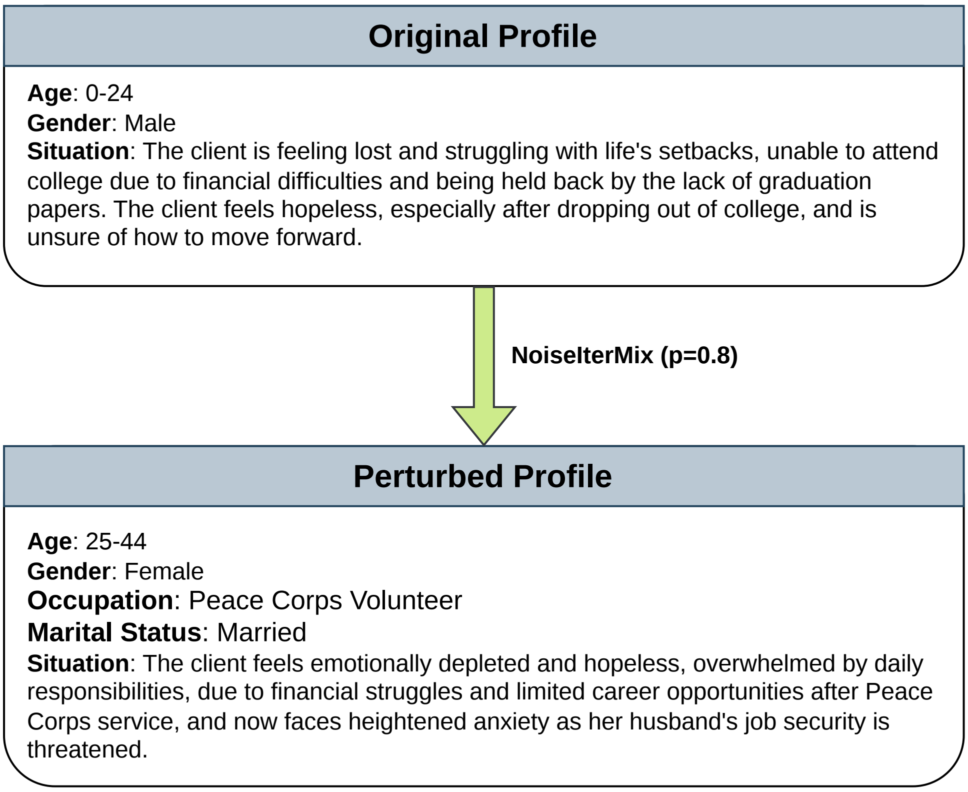

Figure 16: **Example of a perturbed profile under the NoiseIterMix ($p = 0.8$) defense.** Demographic attributes, age and gender are modified, while additional attributes, occupation and marital status, are added. The corresponding situation is adjusted to remain consistent with the perturbed profile while diverging from the original situation.

NER Defense Prompt

System Prompt:
You are a clinical data de-identification specialist.
Your task is to anonymize conversation transcripts by replacing personally identifiable information (PII) with neutral placeholders.
PII to detect and replace:
- Names (people, places that identify a specific person) → [NAME]
- Explicit ages or birth years → [AGE]
- Specific job titles or employer names → [OCCUPATION]
- Marital status (e.g., married, divorced, widowed, single) → [MARITAL_STATUS]
- Explicit gender pronouns or gender statements when the speaker explicitly states their gender → [GENDER]
- Specific company/organization names that could identify a person → [ORGANIZATION]
- Phone numbers, email addresses, addresses → [CONTACT_INFO]
Rules:
- Do NOT change anything else. Preserve the full meaning, emotional tone, and structure of each utterance.
- Do NOT alter generic references (e.g., "my family", "my boss", "a colleague") — only replace if they are specific and identifying.
- Do NOT add commentary or explanation.
- Return the transcript in the exact same format as the input: one turn per line, each prefixed with the speaker role (e.g. "Counselor: ..." / "Client: ...").
- If no PII is found, return the transcript unchanged.
User Prompt:
Anonymize the following counseling session transcript. Return only the transcript lines, no extra text.
{conversation}

Figure 17: Prompt used to perform NER-based redaction on the generated synthetic counseling session.

CoarsProf Situation Adaptation Prompt

System Prompt:
You are a careful text editor working on a mental health research dataset.
STRICT RULE: Make only the minimal edits necessary to remove or generalise identifying information. Do NOT rephrase, restructure, or rewrite sentences that do not contain identifying information. Preserve all original wording that does not need to change.
Your task: make targeted edits to the "situation of the client" description so that it:
1. Removes or generalises potentially identifying information:
   - Personal names → replace with "the client", "a family member", "a friend", etc.
   - Named institutions (schools, companies, hospitals, cities, countries) → replace with generic equivalents like "a university", "a workplace", "a hospital", "a city"
   - Unique personal details that could identify a specific person (e.g. very specific job titles, rare diagnoses combined with unusual life events) → generalise minimally
If none of these are present, return the original text unchanged.
2. Preserves all clinically and emotionally relevant content:
   - The nature of the psychological distress
   - Relationship dynamics and interpersonal conflicts
   - Life circumstances that are contextually important (job loss, bereavement, academic pressure, etc.)
   - The severity and duration of symptoms
3. Keeps the description in third-person, in the same length and style as the original.
Return ONLY the edited text, with no preamble, explanation, or quotation marks.
User Prompt:
{situation}

Figure 18: Prompt used to generalize patient situation in CoarsProf.

NoiseProf Situation Adaptation Prompt

User Prompt:
You are a careful text editor. Your job is to make minimal targeted edits to a client situation description when demographic attributes change.
STRICT RULE: The output must have the same structure and information density as the original. Do NOT add any demographic descriptors (age, gender, occupation, marital status, etc.) that are not already present in the original text. The profile attributes are a background constraint for coherence only — they must NOT appear in the output text unless they were already in the input text.
ORIGINAL situation ({target_words} words):
{situation}
ATTRIBUTE CHANGES:
{attribute_changes}
FINAL PROFILE ATTRIBUTES (coherence check only — do NOT insert these into the text):
{final_attributes}
SYMPTOMS TO KEEP CONSISTENT (do NOT quote severity scores):
{symptoms}
Editing rules:
- Only edit what is directly necessitated by the ATTRIBUTE CHANGES above
- Preserve all original phrasing that does not need to change
- Adapt specific circumstances to fit the new profile (e.g. if occupation changed, replace work-related details with equivalent ones for the new occupation)
- Preserve all clinically and emotionally relevant content: the nature of the psychological distress, relationship dynamics, interpersonal conflicts, and life circumstances that are contextually important
- Do NOT add new sentences or details not present in the original — only adapt existing ones
- Do NOT expand on symptoms, cognitive distortions, or any detail not present in the original
- Keep the description in third-person and at about {target_words} words (do NOT make it longer)
- Do NOT include numerical severity scores
- Return ONLY the edited text, no preamble.

Figure 19: Prompt used to adapt patient situation in NoiseProf.

NoiseIterMix Length Enforcement Prompt

User Prompt:
Condense the following client situation to about {target_words} words (it is currently too long). Preserve the core issue, all clinically relevant content, and third-person voice. Do NOT add anything. Remove only redundancy and padding. Return ONLY the condensed text, no preamble.
TEXT:
{text}

Figure 20: Prompt used to enforce length for patient situation in NoiseIterMix.

NoiseIterMix Situation Divergence Prompt

User Prompt:
You are lightly revising a client situation description for a mental health research dataset. Reword the text below and, where natural, make SMALL substitutions of peripheral details so the text reads differently from an earlier version. The goal is to reduce textual overlap while keeping the client's core issue and emotional experience essentially the same.
CURRENT text ({target_words} words):
{current}
{reference_situation}
FINAL PROFILE ATTRIBUTES (the result must stay coherent with ALL of these; do NOT insert them into the text unless already present):
{final_attributes}
SYMPTOMS TO KEEP CONSISTENT (do NOT quote severity scores):
{symptoms}
Revision rules:
- Keep the client's CORE issue, central distress, and its severity exactly the same
- Peripheral details may be swapped only for very close equivalents (e.g. interest in music -> interest in dance, a sibling -> a cousin) — never for something distant or unrelated
- If using the REFERENCE, take at most ONE small detail or just its style — never its core issue
- Every substitution must stay coherent with the FINAL PROFILE ATTRIBUTES above
- Keep the SAME length, about {target_words} words — do NOT make it longer
- Keep third-person voice; do NOT introduce demographic descriptors not already in the text
- Do NOT include numerical severity scores
- Vary vocabulary and sentence structure from the current text
- Return ONLY the revised text, no preamble.

Figure 21: Prompt used to diverge patient situation in NoiseIterMix iterations.